Event Narrative Module, version 2

Deliverable D5.1.2

Version FINAL

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Abstract: This deliverable describes the results on event modeling achieved during the second year of the NewsReader EU project. Several contributions are presented: (i) a novel two-step “bag of events” approach to perform event coreference, where events are clustered based on event slots such actions, locations, times, human and non-human participants, to form an event-based document summary; (ii) an implementation of the “bag of events” coreference approach as part of the NAF2SEM processing tool, which transforms event mentions (from NAF annotated files) into event instances (in RDF-TRiG format); (iii) two novel approaches for event relation extraction, one for temporal relation extraction and one for causal relation extraction, that enable to relate event instances, and thus set the basis to build storylines; (iv) a novel preliminary method to build timelines (i.e., a storyline formed of chronologically anchored and ordered events related to an entity of interest); (v) a method for extracting and combing event information coming from documents in different languages; and (vi) a preliminary version of the module for determining perspective (i.e., relating a source to the statements about an event) and attribution (i.e., values that define the relation between the source and the contextual statement - e.g., denial/confirmation, certain/uncertain).

Table of Revisions

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0.1	7 Nov 2014	Creation of document with structure	Marco Rospocher	FBK
0.2	15 Dec 2014	Added section on Event Relations	Anne-Lyse Minard, Paramita Mirza	FBK
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0.5	18 Jan 2015	Section for cross-lingual extraction	Piek Vossen	VUA
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1.3	31 January 2015	Check by coordinator	VUA	-
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Executive Summary

This deliverable describes the progress on event modeling for the second cycle of the project, addressing the tasks *T05.1 Event merging and chaining*, *T05.2 Event significance and relevance* and *T05.3 Extraction of narrative graphs*. In details, the contributions presented consist in: (i) a novel two-step "bag of events" approach to perform event coreference, where events are clustered based on event slots such actions, locations, times, human and non-human participants, to form an event-based document summary; (ii) an implementation of the "bag of events" coreference approach as part of the NAF2SEM processing tool, which trasnform event mentions (from NAF annotated files) into event instances (in RDF-TRiG format); (iii) two novel approaches for event relation extraction, one for temporal relation extraction and one for causal relation extraction, that enable to relate event instances, and thus set the basis to build storylines; (iv) a novel preliminary method to build timelines (i.e., a storyline formed of chronologically anchored and ordered events related to an entity of interest); (v) a method for extracting and combing event information coming from documents in different languages; and (vi) a preliminary version of the module for determining perpective (i.e., relating a source to the statements about an event) and attribution (i.e., values that define the relation between the source and the contextual statement - e.g., denial/confirmation, certain/uncertain). A plan for the activities of the third year of the proejct is also discussed.

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1 Introduction

The goal of the NewsReader project[1] is to automatically process massive streams of daily news in 4 different languages to reconstruct longer term story lines of events. For this purpose, we extract events mentioned in news articles, the place and date of their occurrence and who is involved. The basic information is represented per news article in the form of the Natural Language Processing Format (NAF, Fokkens *et al.* (2014)). Representations in NAF are mention-based. Mentions of an event, entity, place or time are not unique. The same instance can be mentioned at different places in a single NAF source or different NAF sources. In NewsReader, we ultimately generate representations for unique instances represented by some URI in RDF. Each instance points back all the mentions in the sources. We developed the Grounded Annotation Framework (GAF, Fokkens *et al.* (2013)), which formally distinguishes between mentions of events and entities in NAF and instances of events and entities in the Simple Event Model (SEM, van Hage *et al.* (2011)) connected through **denotedBy** links between representations. Work Package 5 of the NewsReader project is concerned with the interpretation of mentions as instances and the representation in SEM and GAF that is eventually imported into the KnowledgeStore (Corcoglioniti *et al.* (2013)) that is developed in Work Package 6 of NewsReader. Figure 1 shows the position of Work Package 5 modules in the overall process.

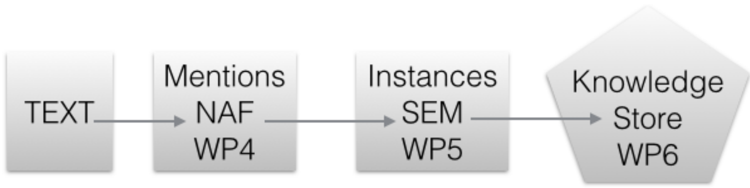


Figure 1: Input-output schema for Work Packages in NewsReader

In this deliverable, we report on the progress of the NAF to SEM/GAF conversion for the second cycle of the project, addressing the tasks *T05.1 Event merging and chaining*, *T05.2 Event significance and relevance* and *T05.3 Extraction of narrative graphs*. In the next section 2, we report on the progress made on event coreference, which is the basis for establishing instance representation, detailing the current implementation of the NAF2SEM module. In section 3, we describe two modules for detecting temporal and causal relations. These relations are used in section 4, where we report on the modules that extract timelines. Timelines are the first step towards creating storylines, for which we outline our plan and current activities.

In Section 5, we provide our first results on cross-lingual event extraction (i.e., combining the event information extracted from documents in different languages). The English

[1] FP7-ICT-316404 *Building structured event indexes of large volumes of financial and economic data for decision making*, www.newsreader-project.eu/

Wikinews corpus was translated to Dutch, Italian and Spanish. By processing the translations with the Dutch and Spanish pipelines, we were able to create SEM representation from each translated data set. These SEM representations should entail the same instance information. We report on the first results of this comparison.

Section 6 discusses a preliminary implementation of the perspective and attribution module. Events are divided into contextual events and source events. The former describe the statements about the changes in the world, while the latter events indicate the relations between sources and these statements. We developed a separate module that derives the perspective and attribution values from these two event types, incorporating the output of the opinion layer and the attribution layer in NAF.

We conclude in Section 7 outlining our activities for the third year of the project.

2 Event Coreference

Event coreference resolution is the task of determining whether two event mentions refer to the same event instance. In this section, we describe a new, robust "bag of events" approach to cross-textual event coreference resolution on news articles. [2]

We first delineate the approach and then describe some experiments with the new method on the ECB+ data set. In the following section, we describe the current implementation within the NewsReader pipeline. The final section reports on the topic classification that will become part of the coreference strategy that we will develop in the 3rd year of the project.

2.1 Bag of Events: a Two-Step Approach

2.1.1 The Overall Approach

It is pretty much common practice to use information coming from event arguments for event coreference resolution (Humphreys *et al.* (1997), Chen and Ji (2009a), Chen and Ji (2009b), Chen *et al.* (2011), Bejan and Harabagiu (2010a), Lee *et al.* (2012), Cybulska and Vossen (2013), Liu *et al.* (2014) amongst others). The research community seems to agree that event context information regarding time and place of an event as well as information about other participants play an important role in resolution of coreference between event mentions. Using entities for event coreference resolution is complicated by the fact, that event descriptions within a sentence often lack pieces of information. As pointed out by Humphreys *et al.* (1997) it could be the case however that a lacking chunk of information might be available elsewhere within discourse borders. News articles, which are the focus of the NewsReader project, can be seen as a form of public discourse (van Dijk (1988)). As such the news follows the Gricean Maxim of quantity (Grice (1975)). Authors do not make their contribution more informative than necessary. This means that information previously communicated within a unit of discourse, unless required, will not be mentioned again. This is a challenge for models comparing separate event mentions with one another on the sentence level. To be able to fully make use of information coming from event arguments, instead of looking at event information available within the same sentence, we propose to take a broader look at event descriptions surrounding the event mention in question within a unit of discourse. For the purpose of this study, we consider a document (a news article) to be our unit of discourse.

We experimented with an **"event template"** approach which employs the structure of event descriptions for event coreference resolution. In the proposed heuristic event mentions are looked at through the perspective of five slots, as annotated in the ECB+ dataset created within the NewsReader project (Cybulska and Vossen (2014b)). The five slots correspond to different elements of event information such as the event trigger (following the ACE (LDC (2005)) terminology) and four kinds of event arguments: time, location, human and non-human participant slots (see Cybulska and Vossen (2014)). The ECB+

[2] The research reported in this section is described in Cybulska and Vossen (2015).

corpus is used in the experiments described here. Our approach determines coreference between descriptions of events through compatibility of slots of the five slot template. The next quote shows an excerpt from topic one, text number seven of the ECB corpus (Bejan and Harabagiu (2010b)).

The "American Pie" actress has entered Promises for undisclosed reasons. The actress, 33, reportedly headed to a Malibu treatment facility on Tuesday.

Consider two event templates presenting the distribution of event information over the five event slots in the two example sentences (tables 1 and 2).

An event template can be created on different levels of information, such as a sentence, a paragraph or an entire document. We propose a novel two step **"bag of events"** approach to event coreference that explicitly employs event- and discourse structure to account for implications of Gricean Maxim of quantity. The approach first fills in an event template per document; lets call it a **"document template"**. By filling in a document template, one creates a "bag of events" for a document, that could be seen as a kind of document **"summary"**. This heuristic employs clues coming from discourse structure and namely those implied by discourse borders. Descriptions of different event mentions occurring within a discourse unit, whether coreferent or related in some other way, unless stated otherwise, tend to share elements of their context. In our example text fragment the first sentence reveals that an actress has entered a rehab facility. From the second sentence the reader finds out where the facility is located (Malibu) and when the "American Pie" actress headed to the treatment center. It is clear to the reader of the example text fragment from the quotation that both event mentions from sentence one and two, happened on Tuesday. Also both sentences mention the same rehab center in Malibu. These observations are crucial for the "bag of events" approach proposed here.

As the first step of the approach a document template is filled, accumulating mentions of the five event slots from a document, as exemplified in table 3. Pairs of document templates are clustered by means of supervised classification. In the second step of the approach coreference is solved between event mentions within document clusters created in step 1. For this task again an event template is filled but this time, it is a **"sentence template"** which per event mention gathers information from the sentence. A supervised classifier solves coreference between pairs of event mentions and finally pairs sharing common mentions are chained into coreference clusters. Figure 2 depicts the implications of the approach for the training data. Figure 3 presents how the test set is processed.

Table 1: Sentence template ECB topic 1, text 7, sentence 1

Action	*entered*
Time	N/A
Location	*Promises*
Human Participant	*actress*
Non-Human Participant	N/A

Table 2: Sentence template ECB topic 1, text 7, sentence 2

Action	*headed*
Time	*on Tuesday*
Location	*to a Malibu treatment facility*
Human Participant	*actress*
Non-Human Participant	N/A

Table 3: Document template ECB topic 1, text 7, sentences 1-2

Action	*entered, headed*
Time	*on Tuesday*
Location	*Promises, to a Malibu treatment facility*
Human Participant	*actress*
Non-Human Participant	N/A

2.1.2 Step 1: Clustering Document Templates Using Bag of Events Features

The first step in this approach is filling in an event template per document. We create a document template by collecting mentions of the five event slots: actions, locations, times, human and non-human participants from a single document. In a document template there is no distinction made between pieces of event information coming from different sentences of a document and no information is kept about elements being part of different mentions. A document template can be seen as a bag of events and event arguments. The template stores unique lemmas, to be precise a set of unique lemmas per event template slot. On the training set of the data, we train a pairwise binary classifier determining whether two document templates share corefering event mentions.

Table 4: ECB+ statistics

ECB+	#
Topics	43
Texts	982
Action mentions	6833
Location mentions	1173
Time mentions	1093
Human participant mentions	4615
Non-human participant mentions	1408
Coreference chains	1958

This is a supervised learning task in which we determine "compatibility" of two document templates if any two mentions from those templates were annotated in the corpus as coreferent. Let m be an event mention, and doc a collection of mentions from a single document template such that $\{m\imath : 1 \leq \imath \leq doc\jmath\}$ where $\imath$ is the index of a mention and $\jmath$ indexes document templates; $doc\jmath : 1 \leq \jmath \leq DOC$ where DOC are all document templates from the corpus. Let ma and mb be mentions from different document templates.

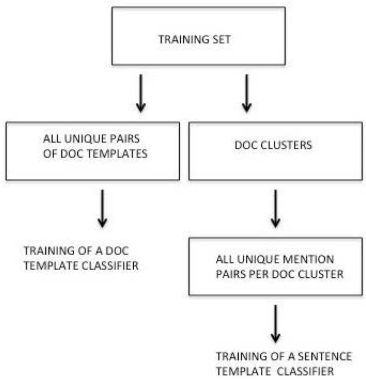


Figure 2: Bag of events approach - training set processing

Figure 3: Bag of events approach - test set processing

"Compatibility" of a pair of document templates (doc_J, doc_{J+1}) is determined based on coreference of any mentions (ma_I, mb_I) from a pair of document templates such that: $coreference(\exists ma_I \in doc_J, \exists mb_I \in doc_{J+1}) \implies compatibility\ (doc_J, doc_{J+1})$.

On the training set we train a binary decision tree classifier (hereafter DT) to find pairs of document templates containing corefering event mentions. After all unique pairs of document templates from the test set have been classified by means of the DT document template classifier, "compatible" pairs are merged into document clusters based on pair overlap.

2.1.3 Step 2: Clustering Sentence Templates

The aim of the second step is to solve coreference between event mentions from document clusters which are the output of the classification task from step 1. We experiment with a supervised decision tree sentence template classifier but this time in the classification task pairs of sentence templates are considered. A sentence template is created for every event action mention annotated in the data set (see examples of sentence templates in table 1 and 2). All possible unique pairs of event mentions (and their sentence templates) are generated within clusters of document templates sharing corefering event mentions in the training set. Pairs of sentence templates that translate into features indicating compatibility across five template slots are used to train a DT sentence template classifier. On the test set; after output clusters of the document template classifier from step 1 are turned to mention pairs (all unique pairs within a document cluster), pairs of sentence templates are classified by means of the sentence template classifier. To identify the final equivalence classes of corefering event mentions, within each document cluster, event mentions are grouped based on corefering pair overlap.

2.1.4 Corpus

For the experiments we used true mentions from the ECB+ corpus (Cybulska and Vossen (2014b)) which is an extended and re-annotated version of the ECB corpus (Bejan and Harabagiu (2010b)). ECB+ is particularly interesting for this experiment because we extended the ECB topics with texts about different event instances but from the same event type (see Cybulska and Vossen (2014)). For example in addition to the earlier mentioned topic of a celebrity checking into a rehab, we added descriptions of another event involving a different celebrity checking into another rehab facility. Likewise, we increased the referential ambiguity for the event mentions. Since the events are similar, we expect that the only way to solve this is through analysis of the event slots. Figure 2.1.4 shows some examples of the seminal events represented in ECB+ with different event instances.

For the experiments on event coreference we used a subset of ECB+ annotations (based on a list of 1840 selected sentences), that were additionally reviewed with focus on coreference relations. Table 4 presents information about the data set used for the experiments. We divided the corpus into a training set (topics 1-35) and test set (topics 36-45).

To-pic	Seminal event type	Human part ECB	Human part ECB+	Time ECB	Time ECB+	Loc ECB	Loc ECB+	Tnr ECB	Tnr ECB+
1	rehab check-in	T.Reid	L.Lohan	2008	2013	Malibu	Rancho Mirage	18	21
2	Oscars host announced	H.Jackman	E.Degeneres	2010	2014	-	-	10	11
3	inmate escape	Brian Nicols,4 dead	A.J. Corneaux Jr.	2008	2009	court-house, Atlanta	prison, Texas	9	11
4	death	B.Page	E.Williams	2008	2013	LA		14	10
5	head coach fired	Philadelphia 76ers, M.Cheeks	Philadelphia 76ers, J.O'Brien	2008	2005	-	-	13	10
6	"Hunger Games" sequel negotiations	C.Weitz	G.Ross	2008	2012	-	-	9	11
7	IBF, IBO, WBO titles de-fended	W.Klitchko, H.Rahman	W.Klitchko, T.Thompson	2008	2012	Germany	Switzer-land	11-1	11
8	explosion at a bank	-	-	2008	2012	Oregon	Athens	8	11
9	ESA changes	Bush	Obama	2008	2009	-	-	10	13
10	eight-year offer	Angels, M.Teixeira	Red Socks, M.Teixeira	2008		-	-	8	13

Figure 4: Overview of seminal events in ECB and ECB+, topics 1-10

2.1.5 Experimental Set Up

The ECB+ texts are available in the XML format. The texts are tokenized, hence no sentence segmentation nor tokenization needed to be done. We POS-tagged (for the purpose of proper verb lemmatization) and lemmatized the corpus sentences. For the experiments we used tools from the Natural Language Toolkit (Bird *et al.* (2009), NLTK version 2.0.4): the NLTK's default POS tagger, Word-Net lemmatizer[3] as well as WordNet synset assignment by the NLTK[4]. For machine learning experiments we used scikit-learn (Pedregosa *et al.* (2011)).

In the experiments different features were assigned values per event slot (see Table 5). The lemma overlap feature (L) expresses a percentage of overlapping lemmas between two instances of an event slot, if instantiated in the sentence (with the exclusion of stop words). Frequently, one ends up with multiple entity mentions from the same sentence for an action mention (the relation between an event and involved entities is not annotated in ECB+). All entity mentions from the sentence are considered. There are two features indicating event mentions' location within discourse (D), specifying if two mentions come from the same sentence and the same document. Action similarity (A) was calculated for a pair of active action mentions using the Leacock and Chodorow measure **?**. Per entity slot (location, time, human and non-human participant) we checked if there are any coreference relations annotated in the corpus between entity mentions from the sentence for the two compared events; we used cosine similarity to express this feature (E). For all five slots a percentage of synset overlap is calculated (S). In case of document templates features referring to active action mentions were disregarded, instead action mentions from a document were considered. All feature values were rounded to the first decimal point.

We experimented with a few feature sets, considering per event slot lemma features only (L), or combining them with other features described in Table 5. Before fed to a

[3]`www.nltk.org/modules/nltk/stem/wordnet.html`

[4]http://nltk.org/_modules/nltk/corpus/reader/wordnet.html

Table 5: Features grouped into four categories: L-Lemma based, A-Action similarity, D-location within Discourse, E-Entity coreference and S-Synset based.

Event Slot	Mentions	Feature Kind	Explanation
Action	Active mentions	Lemma overlap (L) Synset overlap (S) Action similarity (A) Discourse location (D) - document - sentence	Numeric feature: overlap %. Numeric: overlap %. Numeric: Leacock and Chodorow. Binary: - the same document or not. - the same sentence or not.
	Sent. or doc. mentions	Lemma overlap (L) Synset overlap (S)	Numeric: overlap %. Numeric: overlap %.
Location	Sent. or doc mentions	Lemma overlap (L) Entity coreference (E) Synset overlap (S)	Numeric: overlap %. Numeric: cosine similarity. Numeric: overlap %.
Time	Sent. or doc mentions	Lemma overlap (L) Entity coreference (E) Synset overlap (S)	Numeric: overlap %. Numeric: cosine similarity. Numeric: overlap %.
Human Participant	Sent. or doc mentions	Lemma overlap (L) Entity coreference (E) Synset overlap (S)	Numeric: overlap %. Numeric: cosine similarity. Numeric: overlap %.
Non-Human Participant	Sent. or doc mentions	Lemma overlap (L) Entity coreference (E) Synset overlap (S)	Numeric: overlap %. Numeric: cosine similarity. Numeric: overlap %.

classifier, missing values were imputed (no normalization was needed for the scikit-learn DT algorithm). All classifiers were trained on an unbalanced number of pairs of document or sentence templates from the training set. We used grid search with ten fold cross-validation to optimize the hyper-parameters (maximum depth, criterion, minimum samples leafs and split) of the decision-tree algorithm.

2.1.6 Baseline

We will look at two baselines: a singleton baseline and a rule-based lemma match baseline. The singleton baseline considers event coreference evaluation scores generated taking into account all event mentions as singletons. In the singleton baseline response there are no "coreference chains" of more than one element. The rule-based lemma baseline generates event mention coreference clusters based on full overlap between lemma or lemmas of compared event triggers (action slot) from the test set. Table 7 presents baselines' results in terms of recall (R), precision (P) and F-score (F) by employing the coreference resolution evaluation metrics: MUC (Vilain *et al.* (1995)), B3 (Bagga and Baldwin (1998)), CEAF (Luo (2005)), BLANC (Recasens and Hovy (2011)), and CoNLL F1 (Pradhan *et al.* (2011)). When discussing event coreference scores must be noted that some of the commonly used metrics depend on the evaluation data set, with scores going up or down with the number of singleton items in the data Recasens and Hovy (2011). Our singleton baseline gives us zero scores in MUC, which is understandable due to the fact that the MUC measure promotes longer chains. B3 on the other hand seems to give additional points to responses with more singletons, hence the remarkably high scores achieved by the baseline in B3. CEAF and BLANC as well as the CoNLL measures (the latter being an average of MUC, B3 and entity CEAF) give more realistic results. The lemma baseline reaches 62% CoNLL F1. A baseline only considering event triggers, will allow for an interesting comparison with our event template approach, employing event argument features.

2.1.7 Results

Table 6 evaluates the final clusters of corefering event action mentions produced in the experiments by means of the DT algorithm when employing different features. The best coreference evaluation scores with the highest CoNLL F-score of 73% and BLANC F of 72% were reached by the combination of the document template classifier using feature set L across event slots and the sentence template classifier when employing features LDES (see Table 5 for a description of features). Adding action similarity (A) on top of LDES features does not make any difference on decision tree classifiers with a maximum depth of 5. Our best CoNLL F-score of 73% is an 11% improvement over the strong rule based event trigger lemma baseline, and a 34% increase over the singleton baseline.

To quantify the contribution of document templates, we contrast the results of the two-step bag of events approach with scores achieved when skipping step 1 that is without classification of document templates. The results reached with sentence template classification only, give us some insights into the impact of the document template classification

Table 6: Bag of events approach to event coreference resolution, evaluated on the ECB+ in MUC, B3, mention-based CEAF, BLANC and CoNLL F measures.

Step1			Step2			MUC			B3			CEAF	BLANC			CoNLL
Alg	Slot Nr	Features	Alg	Slot Nr	Features	R	P	F	R	P	F	F	R	P	F	F
-	-	-	DT	5	L	61	76	68	66	79	72	61	67	69	68	70
DT	5	L	DT	5	L	71	75	73	71	77	74	64	71	71	71	73
DT	5	L	DT	5	LDES	71	75	73	71	78	74	64	72	71	**72**	**73**
DT	2	L	DT	2	LDES	76	70	73	74	68	71	61	74	68	70	70
DT	5	L	DT	5	LADES	71	75	73	71	78	74	64	72	71	72	73

Table 7: Baseline results on the ECB+: singleton baseline and lemma match of event triggers evaluated in MUC, B3, mention-based CEAF, BLANC and CoNLL F.

Baseline	MUC			B3			CEAF	BLANC			CoNLL
	R	P	F	R	P	F	R/P/F	R	P	F	F
Singleton Baseline	0	0	0	45	100	62	45	50	50	50	39
Action Lemma Baseline	71	60	65	68	58	63	51	65	62	63	**62**

step on our experiment. Note that the sentence template classification without preliminary document template clustering is computationally much more expensive than the two-step template approach, which ultimately takes into account much less item pairs thanks to the initial document template clustering. In the one-step approach, the DT sentence template classifier trained on an unbalanced training set reaches 70% CoNLL F and 68% BLANC F. This is 8% CoNLL F better than the strong baseline disregarding event arguments, but only 3% CoNLL F and BLANC F less than the two-step bag of events approach using feature set L. The reason for the relatively small improvement by the document template classification step could owe to the fact that in the ECB+ corpus not that many sentences are annotated per text. 1840 sentences are annotated in 982 corpus texts, i.e. 1.87 sentence per text. We expect that the impact of document templates would be bigger, if more event descriptions from a discourse unit were taken into account than only the ground truth mentions.

We run an additional experiment with the four entity types bundled into one entity slot. Locations, times, human and non-human participants were combined into a cumulative entity slot resulting in a simplified two-slot template. When using two-slot templates for both, document and sentence classification on the ECB+ 70% CoNLL F score was reached. This is 3% less than with five-slot templates.

To the best of our knowledge, the only related study using clues coming from discourse structure for event coreference resolution was done by Humphreys *et al.* (1997) who perform coreference merging between event template structures. Both approaches determine event compatibility within a discourse representation but we achieve that in a different way, with a much more restricted template (five slots only) which facilitates merging of all event and

Table 8: Best scoring bag of events approach, evaluated in MUC, B3, entity-based CEAF, BLANC and CoNLL F in comparison with related studies. Note that the BOE approach uses gold while related studies system mentions.

Approach	Data	Model	MUC			B3			CEAF	BLANC			CoNLL
			R	P	F	R	P	F	F	R	P	F	F
B&H	ECB B&H 2010	HDp	52	90	66	69	96	80	71	NA	NA	NA	NA
LEE	ECB Lee et al. 2012	LR	63	63	63	63	74	68	34	68	79	72	55
BOE-2	ECB annot. ECB+	DT+DT	65	59	62	77	75	76	72	66	70	67	70
BOE-5	ECB annot. ECB+	DT+DT	64	52	57	76	68	72	68	65	66	65	66
BOE-2	ECB+	DT+DT	76	70	73	74	68	71	67	74	68	70	70
BOE-5	ECB+	DT+DT	71	75	73	71	78	74	71	72	71	72	**73**

entity mentions from a text as the starting point. Humphreys et al. consider discourse events and entities for event coreference resolution while operating on the level of mentions.

Some of the metrics used to score event coreference resolution are dependent on the number of singleton events in the evaluation data set Recasens and Hovy (2011). Hence for the sake of a meaningful comparison it is important to consider similar data sets. The ECB and ECB+ are the only available resources annotated with both: within- and cross-document event coreference. To the best of our knowledge no baseline has been set yet for event coreference resolution on the ECB+ corpus. So in Table 8 we will also look at results achieved on the ECB corpus which is a subset of ECB+, and so the closest to the data set used in our experiments but capturing less ambiguity of the annotated event types Cybulska and Vossen (2014b). We will focus on the CoNLL F measure that was used for comparison of competing coreference resolution systems in the CoNLL 2011 shared task.

The best results of 73% CoNLL F were achieved on the ECB+ by the bag of events approach using five slot event templates (*BOE-5* in Table 8). When using two-slot templates we get 3% less CoNLL F on ECB+. For the sake of comparison, we run an additional experiment on the ECB part of the corpus (annotation by Cybulska and Vossen (2014b)). The ECB was used in related work although with different versions of annotation so not entirely comparable. We run two tests, one with the simplified templates considering two slots only: action and entity slot (as annotated in the ECB by Lee et al. (2012)) and one with five-slot templates. The two slot bag of events (*BOE-2*) on the ECB part of the corpus reached comparable results to related works: 70% CoNLL F, while the five-slot template experiment (*BOE-5*) results in 66% CoNLL F. The approach of Lee et al. (2012) (*LEE*) using linear regression (*LR*) reached 55% CoNLL F although on a much more difficult task entailing event extraction as well. The component similarity method of Cybulska and Vossen (2013) resulted in 70% CoNLL F but on a simpler within topic task (not considered in Table 8). *B&H* in the table refers to the approach of Bejan and Harabagiu (2010) using hierarchical Dirichlet process (in the table referred to by *HDp*); for this study no CoNLL F was reported. In the *BOE* experiments reported in Table 8, during step 1 only lemma features L were used and for sentence template classification (step 2) LDES features were employed. In all tests with the bag of events approach, ground truth mentions were used.

2.1.8 Conclusion

This work presents a two-step bag of events approach to event coreference resolution. Instead of performing topic classification before solving coreference between event mentions, as is done in most studies, the bag of events approach first compares document templates created per discourse unit and only after that, does it compare single event mentions and their arguments. In contrast to a heuristic using a topic classifier, that might have problems distinguishing between different instances of the same event type, the bag of events approach facilitates context disambiguation between event mentions from different discourse units. Grouping events depending on compatibility of event context (time, place and participants) on the discourse level, allows one to take advantage of event context information, which is mentioned only once per unit of discourse and consequently is not always available on the sentence level. From the perspective of performance, the robust bag of events approach using a very small feature set, also significantly restricts the number of compared items. Therefore, it has much lower memory requirements than a pairwise approach operating on the mention level. Given that this approach does not consider any syntactic features and that the evaluation data set is only annotated with 1.8 sentences per text, the evaluation results are highly encouraging.

2.2 From NAF to SEM

The bag of events approach discussed previously has been tested with ground-truth data for events and event arguments. We took the annotated mentions of events, participants, places and time to exclude the impact of errors of the extraction of these mentions in the process. However for a realistic implementation, we need to take the mentions identified by the Event Detection Module as a starting point. So far, we have implemented some of the ideas of the bag-of-event approach in the NewsReader pipeline but further improvements need to follow from future experiments.

2.2.1 Overall architecture

The NAF2SEM processing reads NAF files or streams and results in RDF-TRiG files that contain instances of events, entities and owl:DateTimeDescriptions as well as SEM relations between events, their participants, places, and time. SEM relations are stored as named graphs. The graph URIs are used to store factuality values and provenance relations for each SEM relation based on the GAF model. In Figure 5, an example is shown of the RDF-TRiG that is produced.

The TRiG example 5 shows a graph that includes all the instances. We see here 3 sets of instances: events, entities and time descriptions. In this representation, instances are based on coreference resolution of mentions. Entity mention coreference is established by a the nominal coreference module but also indirectly indirectly by the URI assigned to the entities. Since URIs are unique and most URIs are based on DBpedia, mentions for which we create the same URI are automatically merged and all statements will apply to this

```
@prefix rdfs:  <http://www.w3.org/2000/01/rdf-schema#> .
@prefix time:  <http://www.w3.org/TR/owl-time#> .
@prefix eso:   <http://www.newsreader-project.eu/domain-ontology#> .
@prefix gaf:   <http://groundedannotationframework.org/gaf#> .
@prefix nwrontology: <http://www.newsreader-project.eu/ontologies/> .
@prefix owl:   <http://www.w3.org/2002/07/owl#> .
@prefix rdf:   <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .
@prefix sem:   <http://semanticweb.cs.vu.nl/2009/11/sem/> .
@prefix fn:    <http://www.newsreader-project.eu/ontologies/framenet/> .
@prefix nwrdata: <http://www.newsreader-project.eu/data/> .

<http://www.newsreader-project.eu/instances> {
    #### Entities
    <http://dbpedia.org/resource/Jerry_Yang_(entrepreneur)>
            rdfs:label     "Jerry Yang" ;
            gaf:denotedBy  <http://www.newsreader-project.eu/data/cars/2011/11/04/5460-WPW1-JCDN-004G.xml#char=185,195> .

    <http://dbpedia.org/resource/Trafford>
            rdfs:label     "Old Trafford" , "Trafford" , "Trafford council" ;
            gaf:denotedBy
            <http://www.newsreader-project.eu/data/cars/2011/11/04/545V-MS21-JCF1-12S6.xml#char=1981,1989> ,
            <http://www.newsreader-project.eu/data/cars/2011/11/04/545V-MS21-JCF1-12S6.xml#char=1999,2002> ,
            <http://www.newsreader-project.eu/data/cars/2011/11/04/545V-MS21-JCF1-12S6.xml#char=1981,1997> ,
            <http://www.newsreader-project.eu/data/cars/2011/11/04/545V-MS21-JCF1-12S6.xml#char=1713,1725> ,
            <http://www.newsreader-project.eu/data/cars/2011/11/04/545V-MS21-JCF1-12S6.xml#char=973,985> ,
            <http://www.newsreader-project.eu/data/cars/2011/11/04/5461-W851-DYRS-S25N.xml#char=132,144> ,
            <http://www.newsreader-project.eu/data/cars/2011/11/04/5461-W851-DYRS-S25N.xml#char=2103,2115> .

    <http://www.newsreader-project.eu/data/cars/entities/Miami_Beach_Stage_Door>
            a              nwrontology:ORGANIZATION ;
            rdfs:label     "Miami Beach Stage Door" ;
            gaf:denotedBy  <http://www.newsreader-project.eu/data/cars/2011/11/08/546R-CW91-JCBF-22TB.xml#char=2425,2447> .

     #### Events
    <http://www.newsreader-project.eu/data/cars/2011/11/04/5460-WPW1-JCDN-004G.xml#ev20>
            a              sem:Event ;
            rdfs:label     "resignation" ;
            gaf:denotedBy
            <http://www.newsreader-project.eu/data/cars/2011/11/04/5460-WPW1-JCDN-004G.xml#char=198,209> .

    <http://www.newsreader-project.eu/data/cars/2011/11/04/545V-MS21-JCF1-12S6.xml#ev47>
            a              sem:Event ;
            rdfs:label     "transform" ;
            gaf:denotedBy
            <http://www.newsreader-project.eu/data/cars/2011/11/04/545V-MS21-JCF1-12S6.xml#char=960,972> ,
            <http://www.newsreader-project.eu/data/cars/2011/11/04/545V-MS21-JCF1-12S6.xml#char=2181,2192> .

    <http://www.newsreader-project.eu/data/cars/2011/11/08/546R-CW91-JCBF-22TB.xml#ev65>
            a              sem:Event ;
            rdfs:label     "rehearsal" ;
            gaf:denotedBy  <http://www.newsreader-project.eu/data/cars/2011/11/08/546R-CW91-JCBF-22TB.xml#char=2406,2416> .

    #### Time descriptions
    <http://www.newsreader-project.eu/time/20111104>
            a              time:DateTimeDescription ;
            time:day       "---04"^^<http://www.w3.org/2001/XMLSchema#gDay> ;
            time:month     "--11"^^<http://www.w3.org/2001/XMLSchema#gMonth> ;
            time:unitType  time:unitDay ;
            time:year      "2011"^^<http://www.w3.org/2001/XMLSchema#gYear> .

     <http://www.newsreader-project.eu/data/cars/2011/11/04/545V-MS21-JCF1-12S6.xml#dct>
            a                time:Interval ;
            rdfs:label       "2011-11-04T00:00:00" ;
            gaf:denotedBy    <http://www.newsreader-project.eu/data/cars/2011/11/04/545V-MS21-JCF1-12S6.xml#dctm> ;
            time:inDateTime  <http://www.newsreader-project.eu/time/20111104> .

     <http://www.newsreader-project.eu/data/cars/2011/11/05/5461-W851-DYRS-S25N.xml#tmx1>
            a                time:Interval ;
            rdfs:label       "yesterday" ;
            gaf:denotedBy    <http://www.newsreader-project.eu/data/cars/2011/11/05/5461-W851-DYRS-S25N.xml#char=401,410> ;
            time:inDateTime  <http://www.newsreader-project.eu/time/20111104> .

      <http://www.newsreader-project.eu/data/cars/2011/11/08/546R-CW91-JCBF-22TB.xml#tmx10>
            a                time:Interval ;
            rdfs:label       "Friday" ;
            gaf:denotedBy    <http://www.newsreader-project.eu/data/cars/2011/11/08/546R-CW91-JCBF-22TB.xml#char=2513,2519> ;
            time:inDateTime  <http://www.newsreader-project.eu/time/20111104> .
}
```

Figure 5: Example of instances in TRiG format from the car data set

unique representation across different NAF files. In this example we see that the entity

`<http://dbpedia.org/resource/Trafford>`

has many different mentions originating from two different sources: 545V-MS21-JCF1-12S6.xml and 5461-W851-DYRS-S25N.xml. Entities that are not found in DBpedia are also represented by a URI that we create from their most frequent label, e.g.

`<http://www.newsreader-project.eu/data/cars/entities/Miami_Beach_Stage_Door>`

. The domain of the URI is only unique within the NewsReader data set for cars. This means that the same entity reference across documents will match within the car data set as well.

For each NAF, event coreference is established as well but the URI for a coreference sets of events is created using the source document identifier. This makes event coreference sets unique within each document, i.e. event labels are not automatically matched across documents. For the cross-document event coreference, we use a separate process described for the NAF2SEM module below.

The third type of entity are the time references. We show here 3 examples, one based on the document publication time mention (dctm) and two textual references *yesterday* and *Friday*. These mentions resolve to the same owl-time:DateTimeDescription /textit20111104.

In the next two examples we show how relations and properties are expressed involving the instances. In example 6, we see various actor and time relations for the events. Each SEM relation is a named-graph with a unique identifier based on a mention in the Semantic Role Layer or a time expression. The named graph identifiers are used in example 7 to express where in which sources the relation is expressed and what possible factuality values apply to each statement.

This representation differs only marginally from the represented generated by the first version of the NAF2SEM module that was described in Vossen *et al.* (2013). The main differences between version 1 and version 2 of the NAF2SEM module reside in the way the process has been set up and how the interpretation was achieved. Version 1 of NAF2SEM required a presorting of NAF files by publication date or some other grouping based on meta data. A comparison of NAF based representations of instances and relations was done for all the NAF files in the same folder. This approach relies heavily on the assumption that news published on the same date reports on the same events. The data processed in the first year of the project did not have a layer for time expressions in the document. Consequently, the only way events could be compared was using the document publication time. For version 2, we could exploit the time expresion layer in NAF that provides a detection of time expression in the document and also an interpretation of the time expression to dates, see example 8.

Thanks to the time expression layer in NAF, we could redesign the NAF2SEM module to take any set of NAF files or streams as input to carry out the comparison but also to consider a more fine-grained analysis of events bound to time-points. We thus allow for

```
 ###### SEM relations
<http://www.newsreader-project.eu/data/cars/2011/11/04/5460-WPW1-JCDN-004G.xml#pr6,rl13> {
    <http://www.newsreader-project.eu/data/cars/2011/11/04/5460-WPW1-JCDN-004G.xml#ev20>
            sem:hasActor  <http://dbpedia.org/resource/Jerry_Yang_(entrepreneur)> ;
            <http://www.newsreader-project.eu/ontologies/propbank/A1>
                    <http://dbpedia.org/resource/Jerry_Yang_(entrepreneur)> .
}

<http://www.newsreader-project.eu/data/cars/2011/11/04/545V-MS21-JCF1-12S6.xml#pr32,rl60> {
    <http://www.newsreader-project.eu/data/cars/2011/11/04/545V-MS21-JCF1-12S6.xml#ev47>
            sem:hasActor  <http://dbpedia.org/resource/Trafford> ;
            <http://www.newsreader-project.eu/ontologies/propbank/A1>
                    <http://dbpedia.org/resource/Trafford> .
}

<http://www.newsreader-project.eu/data/cars/2011/11/08/546R-CW91-JCBF-22TB.xml#pr75,rl170> {
    <http://www.newsreader-project.eu/data/cars/2011/11/08/546R-CW91-JCBF-22TB.xml#ev65>
            sem:hasActor  <http://www.newsreader-project.eu/data/cars/entities/Miami_Beach_Stage_Door> ;
            <http://www.newsreader-project.eu/ontologies/propbank/A1>
                    <http://www.newsreader-project.eu/data/cars/entities/Miami_Beach_Stage_Door> .
}

<http://www.newsreader-project.eu/data/cars/2011/11/04/5460-WPW1-JCDN-004G.xml#tr9> {
    <http://www.newsreader-project.eu/data/cars/2011/11/04/5460-WPW1-JCDN-004G.xml#ev20>
            sem:hasTime  <http://www.newsreader-project.eu/data/cars/2011/11/04/5460-WPW1-JCDN-004G.xml#tmx1> .
}

<http://www.newsreader-project.eu/data/cars/2011/11/04/545V-MS21-JCF1-12S6.xml#dt20> {
    <http://www.newsreader-project.eu/data/cars/2011/11/04/545V-MS21-JCF1-12S6.xml#ev47>
            sem:hasTime  <http://www.newsreader-project.eu/data/cars/2011/11/04/545V-MS21-JCF1-12S6.xml#dct> .
}

<http://www.newsreader-project.eu/data/cars/2011/11/08/546R-CW91-JCBF-22TB.xml#tr56> {
    <http://www.newsreader-project.eu/data/cars/2011/11/08/546R-CW91-JCBF-22TB.xml#ev65>
            sem:hasTime  <http://www.newsreader-project.eu/data/cars/2011/11/08/546R-CW91-JCBF-22TB.xml#tmx10> .
}
```

Figure 6: Example of SEM relations in TRiG format from the car data set

```
 #### Factuality of statements
<http://www.newsreader-project.eu/data/cars/2011/11/04/545V-MS21-JCF1-12S6.xml#fv151> {
    <http://www.newsreader-project.eu/data/cars/2011/11/04/545V-MS21-JCF1-12S6.xml#ev47>
            <http://www.newsreader-project.eu/ontologies/value/hasFactBankValue>
                    "CT+" .
}

#### Provenance of statements
<http://www.newsreader-project.eu/data/cars/2011/11/08/546R-CW91-JCBF-22TB.xml#pr75,rl170>
            gaf:denotedBy  <http://www.newsreader-project.eu/data/cars/2011/11/08/546R-CW91-JCBF-22TB.xml#char=2406,2474> .
    <http://www.newsreader-project.eu/data/cars/2011/11/08/546R-CW91-JCBF-22TB.xml#tr56>
            gaf:denotedBy  <http://www.newsreader-project.eu/data/cars/2011/11/08/546R-CW91-JCBF-22TB.xml#char=2406,2519> .
```

Figure 7: Example of Factuality and Provenance statetements in TRiG format from the car data set

```
  <timeExpressions>
    <timex3 id="tmx0" type="DATE" value="2003-01-01"/>
    <timex3 id="tmx1" type="DATE" value="2002-04">
      <!--April-->
      <span>
        <target id="w142"/>
      </span>
    </timex3>
    <timex3 id="tmx2" type="DATE" value="2002-05">
      <!--May.-->
      <span>
        <target id="w148"/>
      </span>
    </timex3>
  </timeExpressions>
```

Figure 8: Time expression layer in NAF

events reported in news on different publication dates to be similar and, the other way around, events reported in the same source to be different based on their time description. The new design also allows for better integration of the module with the KnowledgeStore. In fact, we can compare the events of any NAF file or stream with any set of events retrieved from the KnowledgeStore through any SPARQL request. The new process of the NAF2SEM module is divided into two steps, shown in Figure 9.

1. Create binary object files with all SEM event related data in separate Time Description folders.

2. Compare all object files in the same Time Description folder to establish cross-document event instances and create a single RDF-TRiG file per folder with all the data.

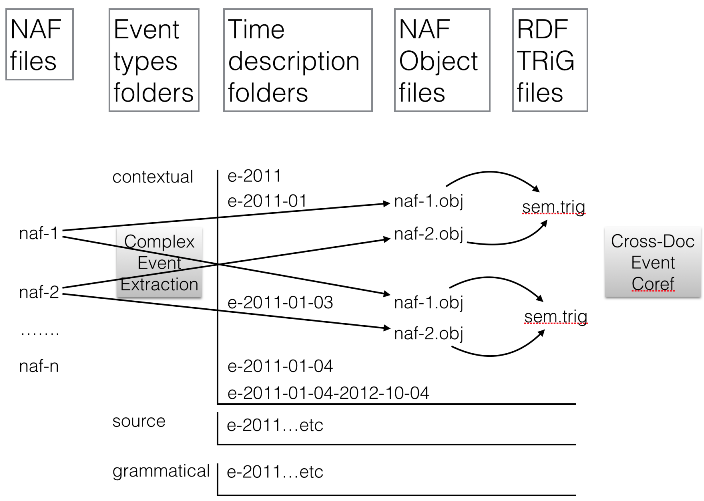


Figure 9: **Overview of the NAF to SEM process**

In the first step, we first obtain all instances as SEM objects with unique URIs and all SEM relations between these objects from a NAF file, in Figure 9 NAF-1 and NAF-2. From these SEM objects and relations, we create so-called CompositeEvent objects for each event instance. A CompositeEvent object contains a single SEM event object and all the other SEM objects and SEM relations related to this SEM event. All event instances are bound to a time description, which can be a year, a month or day or a combination of disjunct times. For each NAF file, we create as many binary object files as there are different time descriptions associated with the events. The object files are stored in event Time Description folders that correspond with the associated time description. In Figure 9, the two NAF files result in object files in the folders e-2011-01 and 2-2011-01-03.

The Time Description Folders are grouped in separate folder making a distinction between the 3 main classes of events distinguished in NewsReader: source-introducing events, grammatical events and contextuals. We use a classification of FrameNet frames to distinguish these different types of events. Events that have no frame as an external reference are considered to be *contextual* by default. Likewise, the result structure consists of 3 main folders: source, contextual and grammatical, with a range of Time Description subfolders and multiple object files within each Time Description Folder.

In the second step, we load all the object files from each subfolder and compare the events to determine cross-document coreference. If two events are corefential, all the information is merged. If not, it is kept separate. Finally, all the SEM objects, SEM relations and GAF relations are serialized to a single RDF-TRiG file per folder.

Since the first step takes NAF streams as input, it can be included into the regular processing pipeline of NewsReader. This process can be parallelized because each NAF file will generate different unique object files. Parallel processing needs to share the Time Description folders or the folders need to be merged in a separate step. The second step needs to process all the objects files in the same folder and can only be parallelized per folder. For processing the car data set, we used parallel processes for both steps.

2.2.2 Cross-document coreference algorithm

We next describe the two steps of the NAF2SEM module in more detail.

Clustering events The first step creates so-called CompositeEventObjects from single NAF files and stores these in corresponding event type and time folders as object files. A CompositeEventObject consists of the SEM event, all its actors as SEM actor, the time descriptions as SEM time objects, the SEM relations between the event and the participants and time descriptions, the factuality of the event and the relations and the attribution information for each mention of a relation. SEM events and SEM actors are internal Java objects based on the (extended) coreference sets of individual NAF files and represented by a unique URI. As a coreference set, they represent all the mentions of an entity and an event in a document or discourse, similar to the bag-of-event approach discussed in subsection **??**.

Main class:

`eu.newsreader.eventcoreference.naf.ClusterEventObjects`

Parameters:

–naf-folder path where the NAF files were found, it recursively scans the folder and all the subfolders

–extension suffix to filter files as input NAF file: files ending with value of extension

–event-folder path to a folder where the output will be stored. The program will create a folder with "events" and the subfolders "source", "grammatical", "contextual"

–project project name that is used to create unique URIs for a data set

–communication-frames path to a file with the FrameNet frames for communication

–contextual-frames path to a file with the FrameNet frames for contextual events

–grammatical-frames path to a file with the FrameNet frames for grammatical events

Example:

```
java -Xmx2000m -cp ../lib/EventCoreference-1.0-SNAPSHOT-jar-with-dependencies.jar
eu.newsreader.eventcoreference.naf.ClusterEventObjects
--naf-folder "/NWR/NWR-DATA/cars-2/1"
--event-folder "/NWR/NWR-DATA/cars-2/1"
--extension ".xml"
--project cars
--communication-frames "../resources/communication.txt"
--grammatical-frames "../resources/grammatical.txt"
--contextual-frames "../resources/contextual.txt"
```

The first step in the algorithm is to read events, entities, time descriptions and relations from the NAF file. For the events and entities, we take the coreference sets from the coreference layer as a starting point.

For entities, we first build up a map for each external reference URI that list all the entities in NAF that have this URI. Next, we iterate over all the entity coreference sets and check the URI-entity map to see which list of entities overlaps most with the coreference sets. Overlap is measured by counting the number of times a set of terms representing a span matches across the coreference set and the entities. Note that the spans in the entity layer and the coreference sets can consist of multiple words and spans are defined differently across NLP modules. For each coreference set, we select the highest matching URI and merge the data of all the associated entities (URI, mentions, phrases, external references) with the data of the coreference set. There may be entities left that did not match with any coreference set. These entities are added as separate entity instances. For each entity instance we create a unique URI, which is either the DBPedia URI if there is one, or a phrase-based URI within the project entity domain, e.g. `http://www.newsreader-project.eu/data/cars/entities/Miami_Beach_Stage_Door`. For all entity instances, we create mention objects based on the source document and the character offsets.

For events, we directly take the event-coreference output as a starting point, excluding all events with a phrase less than 3 characters. We match the span elements in the coreference set with all the predicates from the Semantic Role Layer. If there is an exact match of span elements, we add all data from the predicate (mentions, phrases and external references) to the event instance. We create a single label from the most frequent phrase and finally we create a unique URI within the document. Finally, we check the mentions for each event against the factuality layer in NAF to see if a factuality value is expressed. If so, we add the factuality value to the mention of the event.

For time object, we create an own:Time object from the publication date and for all the time expressions that we find in the Time Expression Layer of NAF. If for some reason, there is no valid normalized time value, we exclude the time object. Invalid time values are empty values of values without a year specified.

After creating the instance objects, we check if events and entities have overlapping spans. Due to the way the MATE tool distinguishes predicates and roles, it can happen that the same tokens are considered both entities and events. If that is the case, we keep the entity reference and remove the event reference.

Next, we extract relations between the instances based on the Semantic Role Layer (SRL) for event-participant relations and the sentence identifiers for event-time relations. We iterate over all the predicates and roles from the SRL. If a predicate is matched with an event object, we check all the roles. Invalid roles are skipped. A role is invalid if it is not of the type PRIMEPARTICIPANT, NONPRIMEPARTICIPANT or LOCATION:

PRIMEPARTICIPANT = "a0","arg0", "a-0", "arg-0";

SECONDPARTICIPANT = "a1","arg1", "a-1", "arg-1";

NONPRIMEPARTICIPANT = "a1", "a2", "a3", "a4", "arg1", "arg2", "arg3", "a-1", "a-2", "a-3", "a-4", "arg-1", "arg-2", "arg-3", "arg-4";

LOCATION = "AM-LOC";

TIME = "AM-TMP";

If valid, we check the span of the role with the spans of all the entity objects that we created before. The matching function counts all the content words of the role span and each entity mention since role spans can include any kind of phrase including complete sentences. Content words need to have a part-of-speech-value "r", "n", "v", "a" or "g". Both the entity mention and the role span should have at least one content word. We count the matching content words across the two spans and calculate the average match of entity to role and role to entity matches given all the content words in each. If the averaged score is above the SPANMATCHTHRESHOLD (set to 75%), we consider that there is a match between the role and the entity. We only consider roles with a match with an entity object for relations. We then create a relation URI based on the document URI, the predicate identifier and the role identifier. We create a mention based on the mention of the predicate and the mention of the role, including all the intermediate tokens. We then use all the external references of the roles the create a set of predicates, the event object URI for the subject and the entity object URI for the object element of the relation triple.

For the event-time relations, we first consider all the time expressions for each event. First we check if the time expression is in the same sentence as the event. If not we check if it is in the previous or next sentence. If not we check if it is in the two preceding sentences. If there is a match, we create a SEM relation with the event URI as subject, the time URI as object and a sem:hasTime predicate relation. We get the mentions of the relation by taking the full span between the predicate and the time expression. The URI of the relation is based on the document URI and a unique tim expression counter per document. If no time relation is created for an event object, we link it to the document publication time. We finally extract factuality statements for each event mention and represent these as separate relations with a unique URI based on the document name and a factuality counter.

After creating the objects and relations, we output so-called CompositeEventObject data to object files as described above. Events can be associated with more than one time expression. If there are too many time expressions (TIMEEXPRESSIONMAX = 5), we ignore the event. Else, we create a sorted set of time-expressions to create a folder name by the temporal sequence of events. Within the division of contextual, source and grammatical events, each time-description folder is unique. We then store each event object, all the related entities, time objects and relations in a corresponding object file.

Matching event objects For the next step, we read the object files (.obj) generated by the ClusterEventObjects program and compare the different events to decide if events across different sources are the same or not. Different types of events will be compared in different ways. The event-type should be set to choose the method of comparison. Matched events represent a single instance and all the information on the events is combined. Non-matched events represent separate instances. The program generates RDF output in TRiG format.

Main class:

```
eu.newsreader.eventcoreference.naf.MatchEventObjects
```

Parameters:

–**event-folder** path to a folder where the input object files are found. Typically, these are the output event folders from the cluster function: will be stored: "events/source", "events/grammatical", "events/contextual".

–**event-type** values are: CONTEXTUAL = "other", "contextual"; COMMUNICATION = "speech", "communication", "cognition", "source"; GRAMMATICAL = "grammatical". This parameter determines how strict event components need to match to establish cross-document event coreference.

–**single-output** (OPTIONAL) if present a separate TRiG file is generated for each object file. If not a single TRiG file is created for the whole folder. Depending on the size and number of object files, you may prefer one or the other. Since the current cluster function creates object files per NAF file, you should always use this parameter for NAF-based object files stored in the same time-description folder.

–**source-data** (OPTIONAL) Path to a file that matches URL for sources with the publisher and or author of the source. If this file is present, an additional provenance layer is created in RDF.

Examples:

```
java -Xmx2000m -cp ../lib/EventCoreference-1.0-SNAPSHOT-jar-with-dependencies.jar
eu.newsreader.eventcoreference.naf.MatchEventObjects
--event-folder "/Users/piek/Desktop/NWR/NWR-DATA/cars-2/1/events/contextual"
--single-output
--event-type "contextual"
--source-data "../resources/LN-coremetadata.txt"
```

Example:

```
java -Xmx2000m -cp ../lib/EventCoreference-1.0-SNAPSHOT-jar-with-dependencies.jar
eu.newsreader.eventcoreference.naf.MatchEventObjects
--event-folder "/Users/piek/Desktop/NWR/NWR-DATA/cars-2/1/events/source"
--single-output
--event-type "source"
--source-data "../LN-coremetadata.txt"
```

Example:

```
java -Xmx2000m -cp ../lib/EventCoreference-1.0-SNAPSHOT-jar-with-dependencies.jar
eu.newsreader.eventcoreference.naf.MatchEventObjects
--event-folder "/Users/piek/Desktop/NWR/NWR-DATA/cars-2/1/events/grammatical"
--single-output
--event-type "grammatical"
--source-data "../resources/LN-coremetadata.txt"
```

This function reads the binary object files from an time-description folder one-by-one and reconstructs the event objects and all the related data as a CompositeEventObject. For each event phrase, we create a list of all the objects. The phrase is based on the most representative lemma from the coreference set from which the event is derived.

We then compare each CompositeEventObject with all the CompositeEventObjects stored in memory with the same phrase to see if there is a match. If there is a match, it merges all the information of the match object with the object in memory. If not, it adds the CompositeEventObject to the storage in memory as a new event.

For comparing the events, we use different heuristics for the type of events. In the case of contextual events at least one of the participating entities should be the same: same URI or the same 'best phrase' as a label. The 'best phrase' is the most frequent phrase that contains alpha-numeric characters and is longer than 2 characters. For source events (speech acts, cognitive events, etc.), a PRIMEPARTICIPANT should match, while for grammatical events (e.g. *Ford launched a new campaign*) there should be a match for both a PRIMEPARTICIPANT and a NONPRIMEPARTICIPANT.

When merging information in case of a match, we keep the URI from the event in memory and add all mentions from the matched object. We also add external references and phrases. Entities and time descriptions are added as they are. In the case of relations, we check if the predicate or object for the relation are different. If so, we add the relation where we replace the subject by the URI of the event object in memory. In all cases, we add the mentions and factuality values of the matched event object.

Events are thus matched on the basis of strict logical comparison. However since the CompositeEventObjects are similar to the event summaries described in section 2.1 and uses NAF coreferences sets based on WordNet Similarity matches in addition to just the lemmas. The matching of events is still robust.

Once the final storage of CompositeEventObjects is completed, the data are stored in RDF-TriG format as explained above. We only use a restricted set so far to create triples. Instances only get subClass relations to FrameNet, ESO, the basic event types *source*, *grammatical* and *contextual* or, in case of entities not matched to DBPedia, their entity classification: PERSON, ORGANIZATION, etc. The DBPedia subclass information is added independently in the KnowledgeStore from the DBPedia information that is extracted for all DBPedia grounded entities. For the actor and place relations, we only accept PropBank, FrameNet and ESO predicates. If the Propbank role "am-loc" is added, we also add a sem:hasPlace predicate relation, otherwise, we add a sem:hasActor relation. These settings have been used to minimize the number of triples. Triples on other relations and sources can be turned on or off as needed.

Finally, we used a mapping of document identifiers to publishers to add attribution information to the provenance graph. This is based on the meta data that is provided by LexisNexis for the car data set. This is a tab-separated file with ten fields:

col 0 = http://www.newsreader-project.eu/2011/1/21/520D-CBF1-F0JP-W4T4.xml

sourceId, col 1 = 520D-CBF1-F0JP-W4T4

dateString col 2 = 2011-01-21

section,col 3 = CARSGUIDE; Pg. 41

owner, col 4 = Herald Sun (Australia)

author, col 5 = James Stanford

location, col 6 =

title, col 7 = Who makes what

copyRight, col 8 = Copyright 2011 Nationwide News Pty Limited All Rights Reserved

nChar, col 9 = 4232

We currently use the author and owner for creating attributedTo relations. For each statement, we extract a separate statement pointing to the offset in the source through a gaf:denotedBy predicate and to the owner and author (if known) through a prov:wasAttributedTo predicate:

```
<http://www.newsreader-project.eu/data/cars/2011/11/04/545T-K6F1-DY8W-C563.xml#pr17,rl36> {
    <http://www.newsreader-project.eu/data/cars/2011/11/04/545T-K6F1-DY8W-C563.xml#ev44>
            sem:hasActor  <http://dbpedia.org/resource/Italy> ;
            <http://www.newsreader-project.eu/ontologies/propbank/A2>
                    <http://dbpedia.org/resource/Italy> .
}

<http://www.newsreader-project.eu/data/cars/2011/11/04/545T-K6F1-DY8W-C563.xml#pr17,rl36>
            gaf:denotedBy  <http://www.newsreader-project.eu/data/cars/2011/11/04/545T-K6F1-DY8W-C563.xml
            #char=708,736> ;
            <http://www.w3.org/ns/prov#wasAttributedTo>
                    <http://www.newsreader-project.eu/provenance/sourceowner/News_Bites_-_Western_Europe_:_
                    Germany> .
```

2.3 Topic classification

Automatic Text Classification involves assigning a text document to a set of pre-defined classes automatically Aggarwal and Zhai (2012). Obviously, these techniques use to be very useful when processing multiple documents and we need to group them according to the set of pre-defined classes. Among the multiple options listed in Deliverable D4-1 *Language Resources and Linguistic Processors*, we selected JEX - JRC Eurovoc Indexer.[5] JEX is a tool for the automatic annotation of multilingual documents with descriptors from the EuroVoc thesaurus[6]. EuroVoc is a multilingual thesaurus used in various European organisations to categorise their documents. Thus, the tool provides a class from EuroVoc and a number of descriptors with their weights.

To process the NewsReader documents, we have developed a new module which uses JEX for the four languages of the project. Our module works in batches at a document level and given a set of NAF documents, it returns them with the topic information. It provides a fine-grained cross-lingual text classification using the Eurovoc lexicon. For instance, given a document from the LN car dataset, the module decides that the document refers to the following topics:

[5] https://ec.europa.eu/jrc/en/language-technologies/jrc-eurovoc-indexer
[6] http://eurovoc.europa.eu/

```
<category code="1361" label="motor vehicle industry" weight="0.11764598369395554 ">
<category code="4654" label="motor vehicle" weight="0.11111113767117804 ">
<category code="5235" label="technical standard" weight="0.10868758914433976 ">
<category code="4261" label="motor car" weight="0.09707126656537715 ">
<category code="2897" label="approximation of laws" weight="0.09687410453032065 ">
<category code="1318" label="Germany" weight="0.09480575306689519 ">
```


The module includes this information in the topic layer of the NAF document.

In a first experiment, we processed 1000 LN car documents with the module to see if the information offered by the tool could be useful in the financial domain. The results obtained seem promising. For instance, looking at the weightiest descriptor of each document, 172 documents correspond to the "motor car" descriptor, 92 to "South Korea", 83 to "motor vehicle industry", 37 to "EU police cooperation" and so on. Considering all the descriptors of each document, 393 documents are classified with "motor car", 307 with "motor vehicle industry", 226 with "motor vehicle", 219 with "anti-dumping duty", 164 with "originating product" and 148 with "road safety".

Although a deeper analysis is necessary, it seems that the information offered by the tools is a valuable source for event-coreference. The topics detected by the module are language independent and they are classified at different levels.

2.4 Future plans

The next steps for improving the software will be based on an error analysis of the benchmarking to improve current precision and recall. In addition, we plan a few more fundamental changes. The current implementation of event coreference and NAF2SEM conversion in NewsReader approximates the bag-of-event approach tested on the ECB+ data set. Event and entity coreference in a single document lumps together event component information that is otherwise spread over multiple sentences throughout the document. However, the bag-of-event approach goes further in that it groups events and participants of many different events in the document to compare it with any other summary. The NewsReader implementation is different in that it follows a strict time description division for the comparison. This is also necessary to be able to handle large amounts of data that need to be compared and integrated quickly. It is not feasible to compare all summaries of news articles to decide on potential candidates for event-coreference. As an alternative, we will integrate the topic-classification in the event-coreference module. In our current implementation, we first divide events into contextual, source and grammatical events and next compare events within the same time description folder. When the topic classification is integrated, we can make subdivision within the contextual events and create time description folders within each topic subdivision for comparison. A topical subdivision makes it possible to adjust the event comparison, e.g. by making it less strict since there is already a topical match.

Another line of research to follow is that we do not just consider isolated events but compare groups of events from NAF files with some bridging relation (see 4). Such an event grouping captures the story of an article in a better way and thus approximates the bag-of-event approach more. Likewise, we can compare micro-stories from articles across sources within topics and time-description folders. Such micro-stories can have partial matches across sources but are expected to give a more reliable co-reference result for the individual events within such a micro-story.

Finally, we will modify the coreference to directly interact with the KnowledgeStore. In that case, we do not create an object file but we keep the composite events of a NAF file or stream in memory. The second step of the processing then consists of querying the KnowledgeStore for events that should potentially be matched with the target CompositeEvent. The results of the SPARQL request will be represented as CompositeEvents in memory in the same way as they are now read from the object files. Comparing the new target event with the given events then can result in either a match or not. If there is a match, the new information is merged with the given information. If there is no match, the event is saved into the KnowledgeStore as a new event.

The new architecture is shown in Figure 10. In this design, we also include a further division in topics that can come from a topic detection module. This means that each event is not only classified by the 3 main event types in NewsReader but also for different topic labels. In Figure 10, we show a subdivision of contextuals in *takeover* events. The CompositeEvent extraction generates a candidate event that is labeled as *contextual-takeover* and bound to the time description *2011-01-03*. A SPARQL request to the KnowledgeStore results in two given events that have the same topic and the same time description. The Crossdocument Event Coreference function now compares the new event with the given events to determine identity. If identical, the new information is merged with one of the given events and the NAF mentions for the event are updated. If not identical, it is considered as a new *takeover* event and added as such to the KnowledgeStore.

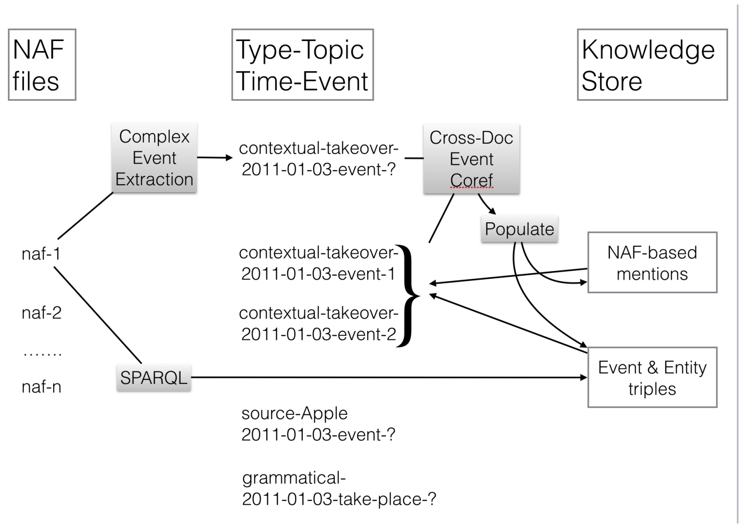


Figure 10: **Overview of the NAF to SEM process with direct integration with the KnowledgeStore**

The new architecture does not require any storage of intermediate result and can be integrated directly into the NewsReader pipeline. Furthermore, we can fine-tune the SPARQL requests to the KnowledgeStore to select the given events for comparison. This can be events for the same topic, same time description and involving the same participants. This new architecture for the NAF2SEM module will be tested for streaming news in relation to a KnowledgeStore with given news to measure the capacity to handle incoming information.

3 Event Relations

Event relation extraction, in particular temporal relation extraction, is a crucial step to anchor an event in time, to build event timelines and to reconstruct the plot of a story.

We have worked on two types of relation: temporal relation and causal relation. In this section we present the event relation representation followed and the approaches used to extract relations in text.

3.1 Annotation Scheme

In TimeML annotation, temporal links are used to *i)* establish the temporal order of two events (*event-event* pair); *ii)* anchor an event to a time expression (*event-timex* pair); and *iii)* establish the temporal order of two time expressions (*timex-timex* pair). Temporal links are annotated with the <TLINK> tag. The full set of temporal relations specified in TimeML version 1.2.1 Saurí *et al.* (2006) contains 14 types of relations, as illustrated in Table 9. Among them there are six paired relations (i.e., with one relation being the inverse of the paired one). These relations map one-to-one to 12 of Allen's 13 basic relations.[7]

Note that according to TimeML 1.2.1 annotation guidelines, the difference between DURING and IS_INCLUDED (also their inverses) is that DURING relation is specified when an event persists throughout a temporal duration (e.g. John *drove* for *5 hours*), while IS_INCLUDED relation is specified when an event happens within a temporal expression (e.g. John *arrived* on *Tuesday*).

a |———|	a is BEFORE b
b |———| (after a ends, with a gap)	b is AFTER a
a |———|	a is IBEFORE b
b |———| (starting where a ends)	b is IAFTER a
a |——|	a BEGINS b
b |—————|	b is BEGUN_BY a
a |——|	a ENDS b
b |—————|	b is ENDED_BY a
a |——|	a is DURING b
b |————————|	b is DURING_INV a
a |————————|	a INCLUDES b
b |——|	b IS_INCLUDED in a
a |———| b |———|	a is SIMULTANEOUS with b
a |————| b	a is IDENTITY with b

Table 9: Temporal relations in TimeML annotation

Similar to the <TLINK> tag in TimeML for temporal relations, we introduce the <CLINK> tag to mark a causal relation between two events. Both TLINKs and CLINKs

[7]Allen's OVERLAPS relations is not represented in TimeML.

mark directional relations, i.e. they involve a source and a target event. However, while a list of relation types is part of the attributes for TLINKs (e.g. Before, After, Includes, etc.), for CLINKs only one relation type is foreseen, going from a *source* (the cause, indicated with $_S$ in the examples) to a *target* (the effect, indicated with $_T$).

We also introduce the notion of causal signals through the <C-SIGNAL> tag. C-SIGNALs are used to mark-up textual elements signalling the presence of causal relations, which include all causal uses of *prepositions* (e.g. because of, as a result of, due to), *conjunctions* (e.g. because, since, so that), *adverbial connectors* (e.g. so, therefore, thus) and *clause-integrated expressions* (e.g. the reason why, the result is, that is why).

Wolff (2007) claims that causation covers three main types of causal concepts, i.e. CAUSE, ENABLE and PREVENT. These causal concepts are lexicalized through three types of verbs listed in Wolff and Song (2003): *i)* CAUSE-type verbs, e.g. *cause, prompt, force*; *ii)* ENABLE-type verbs, e.g. *allow, enable, help*; and *iii)* PREVENT-type verbs, e.g. *block, prevent, restrain.* These categories of causation are taken into account as an attribute of CLINKs.

Given two annotated events, a CLINK is annotated if there is an explicit causal construction linking them. Such construction can be expressed in one of the following ways:

1. Expressions containing **affect verbs** (*affect, influence, determine, change, etc.*), e.g. *Ogun ACN crisis* $_S$ ***influences*** *the launch* $_T$ *of the All Progressive Congress.*

2. Expressions containing **link verbs** (*link, lead, depend on, etc.*), e.g. *An earthquake* $_T$ *in North America was* ***linked*** *to a tsunami* $_S$ *in Japan.*

3. **Basic constructions involving causative verbs** of CAUSE, ENABLE and PREVENT type, e.g. *The purchase* $_S$ ***caused*** *the creation* $_T$ *of the current building.*

4. **Periphrastic constructions involving causative verbs** of CAUSE, ENABLE and PREVENT type, e.g. *The blast* $_S$ ***caused*** *the boat to heel* $_T$ *violently.* With "periphrastic" we mean constructions where a causative verb (*caused*) takes an embedded clause or predicate as a complement expressing a particular result (*heel*).

5. Expressions containing **CSIGNALs**, e.g. *Its shipments declined* $_T$ ***as a result of*** *a reduction* $_S$ *in inventories by service centers.*

3.2 Temporal Relation Extraction

The temporal relation extraction module extracts temporal relations holding between two events or between an event and a time expression. Contrary to TimeML specifications, we are not extracting temporal relations between two time expressions.

Technical details about the implementation the module can be found in the Deliverable 4.2.2 (Section 3.12).

We consider all combinations of event/event and event/timex pairs within the same sentence (in a forward manner) as candidate temporal links. For example, if we have a sentence with entity order such as "...ev_1...ev_2...tmx_1...ev_3...", the candidate pairs are (ev_1, ev_2), (ev_1, tmx_1), (ev_1, ev_3), (ev_2, tmx_1), (ev_2, ev_3) and (ev_3, tmx_1).

The problem of determining the label (i.e. temporal relation type) of a given temporal link can be regarded as a classification problem. Given an ordered pair of entities (e_1, e_2) that could be either event/event or event/timex pair, the classifier has to assign a certain label, namely one of the 14 TimeML temporal relation types.

A classification model is trained for each type of entity pair (event/event and event/timex), as suggested in several previous works ()Mani *et al.* (2006); Chambers (2013)). We build our classification models (Mirza and Tonelli (2014b)) using the Support Vector Machine (SVM) implementation provided by YamCha.[8] The feature vectors built for each pair of entities (e_1, e_2) are as follows:

String and grammatical features. Tokens, lemmas, PoS tags and flat constituent (noun phrase or verbal phrase) of e_1 and e_2, along with a binary feature indicating whether e_1 and e_2 have the same PoS tags (only for event/event pairs).

Textual context. Pair order (only for event/timex pairs, i.e. event/timex or timex/event), textual order (i.e. the appearance order of e_1 and e_2 in the text) and entity distance (i.e. the number of entities occurring between e_1 and e_2).

Entity attributes. Event attributes (*class*, *tense*, *aspect* and *polarity*), and timex *type* attribute[9] of e_1 and e_2 as specified in TimeML annotation. Four binary features are used to represent whether e_1 and e_2 have the same event attributes or not (only for event/event pairs).

Dependency information. Dependency relation type existing between e_1 and e_2, dependency order (i.e. *governor-dependent* or *dependent-governor*), and binary features indicating whether e_1/e_2 is the *root* of the sentence.

Temporal signals. We take into account the list of temporal signals extracted from TimeBank 1.2 corpus. We found that the system performance benefit from distinguishing between event-related signals and timex-related signals, therefore we manually split the signals into two separate lists. Signals such as *when*, *as* and *then* are commonly used to temporally connect events, while signals such as *at*, *for* and *within* more likely occur with time expressions. There are also signals that are used in both cases such as *before*, *after* and *until*, and those kind of signals are added to both lists. Tokens of temporal signals occurring around e_1 and e_2 and and their positions with respect to e_1 and e_2 (i.e. *between* e_1 and e_2, *before* e_1, or at the beginning of the sentence) are used as features.

[8] `http://chasen.org/~taku/software/yamcha/`

[9] The *value* attribute tends to decrease the classifier performance as shown in Mirza and Tonelli (2014b), and therefore, it is excluded from the feature set.

Temporal discourse connectives. Consider the following sentences: *i) "John has been taking that driving course* ***since*** *the accident that took place last week."* and *ii) "John has been taking that driving course* ***since*** *he wants to drive better."* In order to label the temporal link holding between two events, it is important to know whether there are temporal connectives in the surrounding context, because they may contribute in identifying the relation type. For instance, it may be relevant to distinguish whether *since* is used as a temporal or a causal cue (example *i)* and *ii)* resp.). This information about discourse connectives is acquired using the *addDiscourse* tool (Pitler and Nenkova (2009)), which identifies connectives and assigns them to one of four semantic classes in the framework of the Penn Discourse Treebank (The PDTB Research Group (2008)): Temporal, Expansion, Contingency and Comparison. We include as feature whether a discourse connective belonging to the Temporal class occurs in the textual context of e_1 and e_2. Similar to temporal signals, we also include in the feature set the position of the discourse connective with respect to the events.

The result for relation classification (identification of the relation type given the relations) on TempEval3 test corpus is: 58.8% precision, 58.2% recall and 58.5% F1-measure. We compare in Table 10 the performance of *tempRelPro* to the other systems participating in Tempeval-3 task, according to the figures reported in UzZaman *et al.* (2013). *tempRelPro* is the best performing system both in terms of precision and of recall.

System	**F1**	**Precision**	**Recall**
tempRelPro	**58.48%**	**58.80%**	**58.17%**
UTTime-1, 4	56.45%	55.58%	57.35%
UTTime-3, 5	54.70%	53.85%	55.58%
UTTime-2	54.26%	53.20%	55.36%
NavyTime-1	46.83%	46.59%	47.07%
NavyTime-2	43.92%	43.65%	44.20%
JU-CSE	34.77%	35.07%	34.48%

Table 10: Tempeval-3 evaluation on temporal relation classification

3.3 Causal Relation Extraction

In order to extract causal links we need first to detect CSIGNAL in sentences. We implement two different classifiers (Mirza and Tonelli (2014a)). The first one is a CSIGNAL labeler, that takes in as input tokens annotated with information on events and temporal expressions, and classifies whether a token is part of a causal signal or not (Section 3.3.1). The second one is a CLINK classifier, which given an event pair detects whether they are connected by an explicit causal link (Section 3.3.2). Technical details about the implementation of the module can be found in the Deliverable 4.2.2, section 3.13).

3.3.1 Automatic Extraction of CSIGNALs

The task of recognizing CSIGNALs can be seen as a text chunking task, i.e. using a classifier to determine whether a token is part of a causal signal or not. Since the extent of causal signals can be expressed by multi-word expressions, we employ the IOB tagging convention to annotate the data, where each token can either be classified into B-csignal, I-csignal or O (for other). We build our classification model using the Support Vector Machine (SVM) implementation provided by YamCha.[10]

The feature vector for each token includes *token's lemma*, *part-of-speech (PoS) tags*, *flat constituent (noun phrase or verbal phrase)*, *dependency paths*, *named entity type* if the token is part of a named entity, *connective type* if the token is part of a discourse connective, and *binary features* indicating whether a token is part of an event or a temporal expression.

Dependency information is encoded as the dependency relation(s) between the current token and its dependant(s), where the dependant is represented by the first character of its PoS tag. For example, in *"They reject the offer because it was not fully financed"*, there is a dependency relation `SUB (because-PRP, was-VVD)`. Thus, we encode the dependency feature for the token because as `V:SUB`. If there are more than one dependant, the dependency information are concatenated.

The mentioned binary features are introduced to exclude the corresponding token as a candidate token for a causal signal. In other words, if a token is part of an event or a time expression, it is very unlikely that it will be part of a causal signal. The same holds for all connective types that do not express causal relations, e.g. temporal or comparison ones. In order to obtain this information, we include in the feature vector the information about discourse connectives acquired using the *addDiscourse* tool (Pitler and Nenkova (2009)), which is also used in the temporal relation extraction module. Note that causality is part of the Contingency class.

3.3.2 Automatic Extraction of CLINKs

Similar to causal signal extraction, we approach the problem of detecting causal links between events as a supervised classification task. Given an ordered pair or events (e_1,e_2), the classifier has to decide whether there is a causal relation between them or not. However, since we also consider the directionality of the causal link, an event pair (e_1,e_2) is classified into 3 classes: Clink (where e_1 is the source and e_2 is the target), Clink-r (with the reverse order or source and target) or No-rel. Again, we use YamCha to build the classifier. This time, a feature vector is built for each pair of events and not for each token as in the previous classification task.

As candidate event pairs, we take into account every possible combination of events in a sentence in a forward manner. For example, if we have e_1, e_2 and e_3 in a sentence (in this order), the candidate event pairs are (e_1,e_2), (e_1,e_3) and (e_2,e_3). We also include as candidate event pairs the combination of each event in a sentence with events in the following one. This is necessary to account for inter-sentential causality, under the simplifying

[10] `http://chasen.org/~taku/software/yamcha/`

assumption that causality may occur only between events in two consecutive sentences.

We implement a number of features, some of which are computed independently based on either e_1 or e_2, e.g. lemma, PoS, while some others are pairwise features, which are computed based on both elements, e.g. dependency path, signals in between, etc. The implemented features are as follows:

String and grammatical features. The tokens and lemmas of e_1 and e_2, along with their PoS and a binary feature indicating whether e_1 and e_2 have the same PoS tags.

Textual context. The sentence distance and event distance of e_1 and e_2. Sentence distance measures how far e_1 and e_2 are from each other in terms of sentences, i.e. 0 if they are in the same sentence. The event distance corresponds to the number of events occurring between e_1 and e_2 (i.e. if they are adjacent, the distance is 0).

Event attributes. Event attributes as specified in TimeML annotation, which consist of *class*, *tense*, *aspect* and *polarity.* Events being a noun, adjective and preposition do not have tense and aspect attributes in TimeML. Therefore, we retrieve this information by extracting the tense and aspect of the verbs that govern them, based on their dependency relation. We also include four binary features representing whether e_1 and e_2 have the same event attributes or not. These features, especially the *tense* and *aspect* one, are very relevant for detecting causality. For instance, if e_1 is in the future tense and e_2 in the past tense, there cannot be a causal relation connecting e_1 (as source) and e_2 (as target or result).

Dependency information. We include as features *i)* the dependency path that exists between e_1 and e_2, *ii)* the type of causative verb connecting them (if any) and *iii)* binary features indicating whether e_1/e_2 is the *root* of the sentence. Consider the sentence *"Forecasters say the picture will <u>get</u> e_1 worse because more rains <u>are</u> e_2 on the way."* There exists dependency relations `PRP(get, because)` and `SUB(because, are)`. Thus, we encode the dependency path between e_1 and e_2 as `PRP-SUB`.

Causal signals. We take into account the annotated CSIGNALs connecting two candidate events. If there is no annotated CSIGNALs found, we rely on the annotated discourse connectives acquired using the *addDiscourse* tool (Pitler and Nenkova (2009)). We look for causal signals occurring between e_1 and e_2, or before e_1. We also include the position of the signals (*between* or *before*) as feature, since it is crucial to determine the direction of the causality of a given ordered event pair. This is particularly evident if you consider the position of causal signals in the following examples: *i)* "The building <u>collapsed</u> $_{\mathrm{T}}$ **because of** the <u>earthquake</u> $_{\mathrm{S}}$" vs. *ii)* "**Because of** the <u>earthquake</u> $_{\mathrm{S}}$ the building <u>collapsed</u> $_{\mathrm{T}}$." This feature is also very relevant in connection with the *textual context* features, since two events being in two different sentences are linked by an explicit causal relation only in specific cases, for instance if there is a CSIGNAL in between, typically at the beginning of the second sentence. Note that in case of several CSIGNALs occurring between e_1 and e_2, we take the closest CSIGNAL to e_2, as in the sentence "The building was damaged $_S$ **by** the earthquake $_,$ **thus**, people moved $_T$ away". The dependency path between the causal signal and e_1/e_2 is also important to determine the correct involved events in the causal relations. For instance, in the sentence "They decided $_T$ to move **because of** the earthquake $_S$", the involved event is *decided* instead of *move*.

Temporal relations (TLINKs). Rink *et al.* (2010) showed that including temporal relation information in detecting causal links results in improving classification performance. Nevertheless, they only analyze this phenomenon when causality is expressed by the conjunction *and*. We decided to include this information in the feature set by specifying the temporal relation type connecting e_1 and e_2, if any, to see whether TLINKs help in improving causality detection also in a more comprehensive setting. We found that temporal relation types particularly help in disambiguating the direction of causality, i.e. to determine the source and target event.

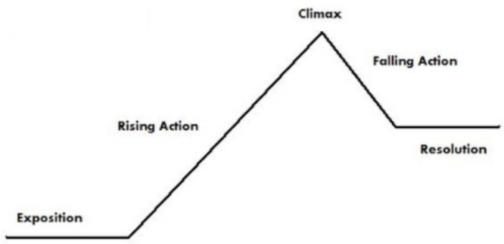
Based on the annotation scheme mentioned in Section 3.1, we manually annotated causality in the TimeBank corpus taken from TempEval-3, containing 183 documents with 6,811 annotated events in total. The total number of annotated CSIGNALs is 171 and there are 318 CLINKs, much less than the number of TLINKs found in the corpus, which is 5,118.

The evaluation is done in a five-fold cross-validation setting, which yields a precision of 67.3%, a recall of 22.6% and an F1 measure of 33.9%.

Since the dataset used is newly constructed, which covers all types of explicit causality between events appearing in the text (the TimeBank corpus), there is no other system to which we can compare our results. Bethard *et al.* (2008) collected 1,000 event pairs connected by the conjunction *and* from the Wall Street Journal corpus. The event pairs were annotated manually with both temporal (BEFORE, AFTER, NO-REL) and causal relations (CAUSE, NO-REL). They use 697 event pairs to train a classification model for causal relations, and use the rest for evaluating the system, which results in 37.4% F-score. Rink *et al.* (2010) perform textual graph classification using the same corpus, and make use of manually annotated temporal relation types as a feature to build a classification model for causal relations between events. This results in 57.9% F-score, 15% improvement in performance compared with the system without the additional feature of temporal relations.

4 Storylines

Stories are seen as the most natural representation of changes over time that also provide explanatory information to these changes. Not all changes make a story. Repetitive changes without further impact, e.g. the rising and dawning of the sun, do not provide a story. We expect that news is typically focused on those changes that have a certain impact. We further assume that the news tries to explain these events (*How did it get so far? Who is responsible?*) and describes the consequences of the event. Our starting point is therefore a key concept from narrative studies, namely that of plot structure (Ryan (1991); Herman *et al.* (2010)). A plot structure is a representational framework which underlies the way a narrative is presented to a receiver. Figure 4 shows a a graphical representation of a plot structure.



We therefore seek to create structures of event selected from all extracted events that approximate such abstract plot structures. Contrary to what can be done in narrative studies, where the documents themselves normally provide a linear development of the plot structure, we aim at obtaining the plot structure from collections of news articles on the same topic and spanning over a period of time. We aim at identifying first the "climax", which in our perspective will correspond to the most salient event in a news article. After the most salient event and its participants have been identified we will use event relations to identify the rising actions (i.e. *how and why did the most salient event occur?*), if any[11], and the falling actions and consequences (i.e. *what happened after the climax? what are the speculations linked to the climax event? ...*). As a first step towards creating a storyline, we need to establish the temporal ordering of events. In subsection 4.1, we describe a first module for extracting such timelines that participated in the SemEval-2015 task. In subsections 4.2, we describe the plans for the 3rd year to move beyond timelines towards plot-based storylines.

4.1 Timeline module

This section reports the first development and evaluation of the Timeline extraction module. This module has been developed on top of the modules developed in the second

[11]Notice that (unforeseeable) natural events, like earthquakes, are to be considered as self-contained climax events.

development cycle of the project and in conduction with the Spinoza Project "Stories and World Views as a Key to Understanding Language" (NWO-Spinoza-2013).

Timeline extraction is a pilot task in Natural Language Processing which aims at reconstructing the chronological temporal order of events form (large) collections of news spanning over different years. In the following subsection, we describe the current implementation within the NewsReader pipeline and the evaluation with respect to a public dataset made available in the context of the SemEval 2015 evaluation exercise Task 4: TimeLine: Cross-Document Event Ordering.

The starting point for the Timeline module is the SEM-TRIG representation format (i.e. instance level). Nevertheless, to address the task and recover relevant information, the Timeline module has to get back to the NAF representation (i.e. mention level) at document level as provided by the different NewsReader pipeline layers in order to get more information that is not in the SEM-TRIG representation. Apart from the basic layers (part of speech tagging, lemmatisation and syntactic parsing), the most relevant layers used to collect and enrich data for the Timeline extraction are:

- named entity detection and disambiguation (NERD layer);
- semantic role labelling (SRL layer);
- nominal and event coreference layer (COREF layer);

Each annotation layer corresponds to a different subtask for the extraction of Timelines. Two additional subtasks, temporal relation and event extraction and classification, have been realised in two different ways: i.) the first, by using the output of the temporal relation (TLINKs) and causal relation (CLINKs) layers in conjunction with the semantic role labelling layer (SRL); ii.) the second, by using the output of a temporal processing tool, TIPSem Llorens *et al.* (2010).

Finally, we developed a dedicated and new rule-based module of analysis, the Timeline layer (TML layer), which will be integrated in the NewsReader pipeline by cycle 3 in the 3rd year (it now runs separately and the output is not integrated in SEM-TRIG). In particular, we are referring to the extension and development of modules in the NewsReader pipeline that extract information at mention level, such as bridging relations between events, relevant for storyline extraction. This information will be added to SEM-TRIG representations as relations between events and made available through the Knowledge Store. Further integration within NewsReader can follow two different lines:

- We build relations on top of the temporal and causal relations between events that represent scripts extending documents and store these as relations in the RDF-TRIG representation.
- We investigate how stories can be created as a service through querying the KnowledgeStore. This may assume that story building blocks are already present in the data structures.

Both approaches are not incompatible with each other.

4.1.1 Timelines: task description

The Task 4: TimeLine: Cross-Document Event Ordering has been proposed as a new pilot task for the SemEval 2015 Evaluation Exercise. The task builds on previous temporal processing tasks organised in previous SemEval editions (TempEval-1[12], TempEval-2[13] and TempEval-3[14]). The task aims at advancing temporal processing by tackling for the first time with a common and public dataset the following issues:

- cross-document and cross-temporal event detection and ordering; and
- entity-based temporal processing.

Following the task guidelines, a Timeline can be defined as a set of chronologically anchored and ordered events related to an entity of interest (i.e. a person, a commercial or financial product, an organization, and similar) obtained from a document collection spanning over a (large) period of time. Furthermore, not all events are eligible to enter a Timeline. The task organisers have restricted the event mentions to specific parts-of-speech and classes, as defined in the task Annotation Guidelines[15], in particular, an event can enter into a Timeline only if the following conditions apply:

- it is realised by a verb, a noun or a pronoun (anaphoric reference);
- it semantically denotes the happening of something (e.g. the launch of a new product) or it describes the action of declaring something, narrating an event, informing about an event;
- it is a factual or certain event, i.e. something which happened in the past or in the present, or for which there is evidence that will happen in the future.

No training data was provided. Only a trial dataset of 30 articles manually annotated from WikiNews, and associated Timelines for six entities, have been provided to the task participants. Each event in the Timeline is associated with a time anchor of type "DATE" following the TimeML Annotation Guidelines (Saurí *et al.* (2006)). In case an event cannot be associated with a specific time anchor, an underspecified time anchor of the type "XXXX-XX-XX" is provided.

The final Timeline representation is composed by a tab field file containing three fields. The first field (ordering) contains a cardinal number which indicates the position of the event in the Timeline. Simultaneous, but not coreferential, events are associated with the same ordering number. Events which cannot be reliably ordered, either because of a missing time anchor or underspecified temporal relations (e.g. an event which is associated

[12]http://www.timeml.org/tempeval/

[13]http://timeml.org/tempeval2/

[14]http://www.cs.york.ac.uk/semeval-2013/task1/

[15]`http://alt.qcri.org/semeval2015/task4/data/uploads/documentation/manualannotationguidelines-task4.pdf`

with a generic date with value "PAST_REF") are to be put at the beginning of the Timeline and associated with cardinal number 0. The second field (time_anchor) contains the time anchor. The third column (event) consists of one event or a list of coreferential events. Each event must be represented by the file id, the sentence id and the extent of the event mention (i.e. the token). To clarify the representation of a timeline, in Figure 4.1.1 we report the output of SPINOZA_VU_1 system for the target entity "*Airbus*"[16].

```
0	XXXX-XX-XX	100911-25-award
0	XXXX-XX-XX	100911-26-said
0	XXXX-XX-XX	102977-0-wins
0	XXXX-XX-XX	102977-10-inked
0	XXXX-XX-XX	129865-4-said
0	XXXX-XX-XX	129865-3-crashed
0	XXXX-XX-XX	145930-8-release
0	XXXX-XX-XX	3307-6-confirmed
0	XXXX-XX-XX	62929-14-announced
0	XXXX-XX-XX	71426-3-take
0	XXXX-XX-XX	71526-16-built
0	XXXX-XX-XX	82597-3-greeted
1	XXXX-XX-XX	3307-5-stated
1	XXXX-XX-XX	82597-2-landed
2	2004-11-24	1173-28-provides
3	2005-04-27	8951-12-sold
4	2005-06-13	11716-0-wins
5	2006-10-05	51682-0-compensated
6	2006-10-05	51682-2-says
7	2006-11-05	25501-3-announced
7	2006-11-05	25501-0-delay
8	2007-06-09	71426-4-exercise
9	2007-06-09	71426-3-said
10	2007-07-09	73808-17-get
11	2007-09-01	78496-2-delayed 78496-0-delayed
12	2007-10-15	82548-2-delivered
13	2007-12-20	87805-0-tells
14	2008-03-01	100911-25-decision
15	2009-XX-XX	129865-3-killing
16	2009-01-XX	149731-2-confirmed
17	2009-06-18	128016-3-signed
18	2009-06-18	128016-0-purchase
18	2009-06-18	61389-3-comes
19	2010-01-XX	149731-10-asked
20	2013-XX-XX	61389-2-announced
21	2023-XX-XX	100911-2-supply
```

Two tracks were proposed. Track A aimed at extracting Timelines for target entities from raw text. Track B aimed at extracting Timelines for target entities by providing manually annotated data for the gold events. Both tracks have a subtrack, Subtrack A and Subtrack B, whose goals are to evaluate only the ordering of events, without taking into account the time anchoring.

The test data provided by the organisers consisted of three different corpora, each containing 30 articles, and an overall 37 target entities (12 for the first corpus, 12 for the second corpus and 13 for the third corpus). The evaluation is based on the TempEval-3 evaluation (UzZaman *et al.* (2012)) tool. All events associated with cardinal number 0 in a Timeline are excluded from the evaluation. Results and ranking reports the micro average F1 score for temporal awareness.

4.1.2 System Description and Evaluation

Two different version of the system for Timeline extraction, called SPINOZA_VU_1 and SPINOZA_VU_2 respectively, have been developed. The main difference between the two

[16] The entity "*Airbus*" was one of the entity provided by the SemEval task organisers

versions, as already stated, concerns the provenance of the information for the identification of events, temporal expressions and temporal relations. Extracting cross-document Timelines from a corpus involves a number of subtasks. We will report here each subtask and the method(s) used for their realisation.

Event extraction The SPINOZA_VU_1 system uses an external tool for the identification of events, namely TIPSem. TIPSem provides as output a TimeML compliant XML document which requires further processing. In order to use the detected events from TIPSem for the Timeline construction, we need first to retain only the eligible events as provided by the task organisers. These are identified by means of a set of post-processing rules which convert and map the TimeML event classes (OCCURRENCE, STATE, I_ACTION, I_STATE, ASPECTUAL, REPORTING and PERCEPTION) into specific FrameNet frames and/or Event Situation Ontology (ESO) types which meet the conditions for events described in the task. Practically, we could immediately retain as eligible events all event mentions classified as REPORTING and PERCEPTION and exclude those classified as ASPECTUAL. The remaining event classes, namely I_ACTION, I_STATE, STATE and OCCURRENCE, had to be filtered. The filtering take place in two steps:

- first, we have aligned the output of TIPSem with the Word Sense Disambiguation (WSD) annotation layer of the NewsReader pipeline. Only the sense with highest score was retained as correct. Sense disambiguation is performed with respect to WordNet (WN) 3.0, i.e. by associating a synset id to the eligible event token;
- the sense disambiguated event token, was then used to access the Predicate Matrix (version 1.1) (Lacalle *et al.* (2014)). The Predicate Matrix is a new lexicon resource which extends the SemLink initiative (Palmer (2009)) by providing useful mapping and integration of multiple sources of predicate information including FrameNet, VerbNet, PropBank, WordNet, and the newly created Event Situation Ontology (Segers *et al.* (2015)).

As for the SPINOZA_VU_2, we used the semantic role annotation layer (SRL layer) of the NewsReader pipeline to identify both the event word candidates and then to filter and retain the eligible events. In this case the access to the Predicate Matrix was not necessary as each predicate in the SRL layer is also associated with corresponding FrameNet frames and ESO types. Thus, only the predicates matching those specific FrameNet frames and/or ESO types were retained as good.

Finally, in order to meet the conditions on events required by the task organizers an additional filter on event factuality has been developed. The factuality filter is an additional rule-based tool which uses syntactic information between the target event(s) and its modifiers. The rules have been developed on the basis of the output of the dependency relation layer from the NewsReader pipeline (DEPREL layer) and targets dependency relation between the event and negative and epistemic adverbs (e.g. possibly), and modal auxiliaries. This means that if a target event is directly modified by or in the scope of a negative adverb, the event is excluded from the set of good events.

Coreference relations Two levels of coreference have to be taken into account: in-document and cross-document. As for the in-document level, both version of the system uses the coreference layer (COREF layer) of the NewsReader pipeline. This layer has been used both to extend the event mentions (identify coreferential events with same lemmas or pronouns) and to identify all mentions of entities. Concerning the cross-document level, two different strategies have been implemented, provided the different nature of the linguistic elements involved in the coreferential relations. In particular:

- cross-document entity mentions was implemented by using the URI links associated to entity mentions. The URIs can correspond either to DBpedia entries or, in case no match is found, to specific URI composed by the entity tokens and a dedicated prefix (e.g. `nwr_wikinews_entity:Northrop+Grumman+and+EADS`[17]). All entity mentions sharing the same URIs from different documents have been associated with the same entity instance, represented by the URI.
- cross-document event coreference has been implemented as a post-processing step of the Timeline creation. The working hypothesis is that two event lemma mentions denote the same event instance, i.e. they co-refer, if they share the same participants, time of occurrence and (possibly) location. We have thus used the information from the preliminary entity-based Timelines, i.e. event lemmas and time anchor, in order to identify instances of cross-document event coreferential expressions. In this preliminary implementation of cross-document event coreference we did not extend the search also to cross-part of speech elements (e.g. *fly* - verb and flight - noun).

Entity identification and linking Entity identification relied mainly on the entity detection and disambiguation layer of the NewsReader pipeline (NERD layer). As already stated, each entity in NewsReader is associated with a unique URIs - either from DBpedia or specifically created. However, the entity extraction for the Timelines has been performed by merging together the information from the NERD layer with that from the SRL layer. After we have identified all eligible event mentions, for each event we extracted its associated argument structure from the SRL layer. Only those events whose target entities fulfill the argument position of proto-agent or proto-patient (Arg0 and Arg1, respectively, in PropBank terminology) have been retained as good. Finally, two strategies have been implemented to retrieve the target entities. The first, search for a perfect match between the target entities provided by the task organisers and the DBpedia URIs. The second approach is based on the Levenshtein distance (Levenshtein (1966)) between the target entities and the tokens of the specific URI. We set an empirical threshold on the basis of experiments conducted on the trial data in order to maximise the precision.

Temporal Relations This layer of analysis is the most important and, at the same time, the most critical one. For the SPINOZA_VU_1 version, we used the Temporal Relation information provided by TIPSem (TLINKs), including temporal expression detection and normalisation. For the SPINOZA_VU_2 version, we have used the TLINK and CLINK

[17]nwr_wikinews_entity prefix stands for http://www.newsreader-project.eu/data/wikinews/entities/

layers of the NewsReader pipeline. As for the CLINK layer, we have converted all causal relations into temporal ones, with value `BEFORE`. For both versions of the system, the following strategies have been applied in order to maximise the set of available temporal relations:

- in case of a TLINK between a target event and a temporal expression classified as `DURATION`, we identified the beginning or ending point of the temporal expression of type `DURATION`. As nor TIPSem nor the NewsReader temporal expression recognition and normalisation layer provide the temporal anchor used for the normalisation of `DURATION`s, we have assumed the Document Creation Time (DCT) as the temporal anchor;
- all TLINKs between a temporal expression and a target event whose value is other than "IS_INCLUDED" have been transformed into anchoring relations;
- all TLINKs involving pairs of target events per entity have been retained as good and used to reconstruct the cross-document Timelines.

Timeline Extraction The Timeline extraction module is responsible for harmonising and ordering cross-document temporal relations (anchoring and ordering). Its implementation relies on a model for cross-document event ordering which provides a method for selecting the initial (relevant) temporal relations (namely, all anchoring relations) and how to update the information so as additional temporal relations (both anchoring and ordering relations) can be inferred. The model is event-based and aims at building a global Timeline between all events and temporal expressions regardless of the entities. Only after the global Timeline has been constructed, we have extracted the entity based Timeline for the target entities.

Our systems took part only to the Track A (both main and subtask) of the SemEval 2015 Task 4. In Table 11 we report the results of both versions of the system for the Task A - Main, including the best performing system. Table 12 reports the results of both versions of the system for the Task A - Subtask. In Table 12 no other result is reported because only our system participated. The F1-scores ranges from 0 to 100.

Table 11: System Results (micro F1 score) for the SemEval 2015 Task 4 Task A - Main

System Version	Corpus 1	Corpus 2	Corpus 3	Overall
SPINOZA_VU_1	4.07	5.31	0.42	3.15
SPINOZA_VU_2	2.67	0.62	0.00	1.05
WHUNLP_1	8.31	6.01	6.86	7.28

Overall, the results are not satisfying. First, out of 37 entity based Timelines, we had been able to obtain results only 31 of them. Three sources of errors are involved in this task and both versions of our system are apparently affected by all of them. A preliminary error analysis has shown that biggest source of errors comes from the connections among

Table 12: System Results (micro F1 score) for the SemEval 2015 Task 4 Task A - Subtask

System Version	Corpus 1	Corpus 2	Corpus 3	Overall
SPINOZA_VU_1	1.20	1.70	2.08	1.69
SPINOZA_VU_2	0.00	0.92	0.00	0.27

the event detection, temporal relation and and semantic role layers. This means that: i.) we may be able to identify the correct event with respect to the target entity, but we are lacking the temporal relation information for that event, thus failing to put it into the Timeline, ii.) we fail in the identification of the target entity as an argument of an event. Nevertheless, the connection between the event identification and the temporal relation information qualifies as the main source of error. The lower performance in Subtask A of both versions of the system is a further cue in support of this analysis. In the Task A - Main Task, each Timeline was evaluated for the correctness with respect to both event anchoring and event ordering. On the contrary, in Task A - Subtask, only event ordering is evaluated. For Task A - Subtask, we submitted the same results obtained for Task A - Main Task with the exception of the time anchors. The big drop in results can only be explained by means of a failure in selecting the events with relevant temporal information. Furthermore, a preliminary analysis conducted on corpus 1 concerning only the event identification (regardless of the event anchoring and ordering) shows that on 11 extracted Timelines we have an average precision on event detection of 30.49% and an average recall of only 12.37% for version 1 of the system. Things are not better on version 2, where average precision is higher, 35.71%, but the average recall is even lower, i.e. only 8.06% .

The difference in performance between the two versions of the systems, SPINOZA_VU_1 and SPINOZA_VU_2, further supports the previous analysis. In particular, the use of an external tool such as TIPSem for event detection and temporal relation proves to be more robust than the use of the TLINKs and CLINKs layers of the NewsReader pipeline. Nevertheless, both versions (and corresponding tools and layers) require improvements. In particular, it is necessary to develop a new TLINK classifier which aims at ordering and anchoring not all events in a document but only those which are relevant (either because they involve a target entity or for saliency-based analyses). Finally, it should be noted that we did not use a baseline rule for ordering events without any time anchor. We could have used to order of mentions in each document to obtain a default relation when there is no explicit relation detected. This will be investigated in a future update of the system.

4.2 From Timelines to Storylines in News

Storylines are usually associated with fictitious (written or oral) texts. A storyline may have different meanings, but it is normally associated with the plot (or subplot) of a (fictional) story. Most works on storylines have been conducted in Narratology and have focused on “closed world” texts (i.e. texts which self-contain all the relevant information) such as short novels, children stories and similar.

From a broad perspective, efforts in the Natural Language Processing (NLP) community for storylines mainly concentrate on three aspects: i.) automatic summarisation (both single document and multi document); ii.) computational models for narrative (i.e. fictional) texts[18] and on tools for the identification of plot units or story grammars (Garnham (1983); Brewer and Lichtenstein (1980); Lehnert (1981); Goyal *et al.* (2010)); and ii.) temporal processing, mainly focusing on the identification of the chronological order of events. Concerning this latter point, language resources, such as annotated corpora (e.g. the TimeBank corpora for English, Spanish, Italian, French, Korean, Portuguese, Rumanian), dedicated annotation schemes (e.g. TimeML and ISO-TimeML) and systems have been developed and evaluated. Much of this work is limited to single documents, i.e. reconstructing the temporal ordering of events in a single document. However, works on cross-document identification of event ordering have been conducted with varying results (Xu *et al.* (2013); Huang and Huang (2013); Nguyen *et al.* (2014); Zhang and Weld (2013)). The common representation of the outcome of such systems is a line with temporal expressions on it where events are anchored and ordered. In a word, these systems produce Timelines. The SemEval 2015 Task 4 made available for the first time a common data set and an evaluation methodology for Timelines.

We aim at a more challenging framework by shifting from Timelines to Storylines. Extracting and identifying a storyline, or plot, requires moving forward from current approaches on temporal processing. In particular, next to the temporal and causal relations between events, it is necessary to:

- understand the goals of the characters (i.e. people, organisations, products etc.) involved in the story;
- understand the motivations of the characters;
- understand and identify the emotional outcomes of the characters;
- understand and identify the relationships between the characters;
- identify and connect events one another by means of new relations (*bridging events*).

Current activities focus on two aspects:

- the development of a more robust temporal relations extraction system;
- experiments on the identification of the most salient events and their participants in a text.

For selecting the relevant events, we will further develop the notion of microstories. We assume that each news article contains one or a few main contextual events, e.g. captured by the title or first paragraph of the news article. Next, we will use bridging relations to connect other events (secondary events) to the main events (primary events). Bridging relations to be considered are:

[18] Since 2009 a dedicated workshop on Computational Models of Narrative (CMN) has been regularly held.

1. Co-participation: entities involved in the primary events are also involved in other secondary events;
2. Temporal relations: secondary events have an explicit temporal relation to a primary event;
3. Causal relations:
   (a) secondary events that have an explicit causal relation to a primary event;
   (b) secondary events that can be related though background script relations, such as FrameNet frame relations;

After extracting microstories from single news-articles, we want to combine the microstories from articles published over time into longer macrostories. This solution will be submitted to the workshop, "Computing News Storylines"[19], which will be held on July 31 in conjunction with ACL-IJCNLP 2015. The goal of the workshop is to gather together, for the first time, researchers and scientists working on narrative extraction and representation from news within Computational Linguistics and Artificial Intelligence. One innovative aspect of the workshop is the re-use the SemEval 2015 Task 4: Timelines dataset from possible participants in order to provide their own annotations, interpretations, and system results. The data will be collected before the workshop and summarized to facilitate an insightful comparison. The results of the combined manual and automatic annotation of the common dataset will be used to dive the discussion around three themes:

- Definitions: what is a storyline? How can it be formally and computationally formulated?
- Resources: What are the core markables of a storyline? How annotation of storylines should be performed? Can existing annotation schemes be re-used and adapted for storyline annotation? How should we annotate cross-document information concerning events and character perspectives? Is it feasible to develop a StoryBank for evaluation?
- Evaluation: How do we determine if an extracted storyline is "good enough"? Can standard measures — such as Precision, Recall and F-measure — be applied to evaluate storyline extraction or do we need different measures? Should evaluation take place at a global level or must it be conducted separately on the different components of a storyline system?

[19]https://sites.google.com/site/computingnewsstorylines2015/home

5 Cross-lingual extraction

The processing of text in NewsReader takes place in two steps. First, language-specific processors interpret text in terms of mentions of entities, events, time-expressions and relations between them. Secondly, we create a language-independent representation of the instances of entities, events, time-expressions and their relations in RDF-TRiG. Although the first step is specific for each language, the representation of the output in NAF is the same for all the 4 NewsReader languages. Differences are restricted to the tokens that make up these mentions. In the case of entities and time-expressions, we normalize these tokens by using the DBpedia URIs or the normalized time for time-expressions. Events are however still represented by the tokens of the predicates. To make the events interoperable across languages, we use the GlobalWordnetGrid to map each predicate to concepts in the InterLingualIndex (ILI). In Figures 11 and 12, we show two fragments in English and Spanish respectively for the same Wikinews article with the representation of an entity, a predicate with roles and a time-expression. The English representation of the entity *Boeing 787* has an external reference to the English DBpedia, where the Spanish entity maps to both the Spanish and English DBpedia entries. By taking the English reference for Spanish, we can map both entity references to each other across the documents. We can see the same for the time-expressions in both languages that are normalized to the same values: *2005-01-29*. In case of the predicates, both examples show external references to various ontologies among which FrameNet and ESO and also to WordNet ILI-concepts. The predicates of these two examples can therefore be mapped into each other through the reference *ili-30-02207206-v*. A similar example is shown in Figure 13 for Dutch NAF where the verb *kopen* is mapped to the same ILI record.

To compare the capacity of the different language pipelines to extract the same information, we use the NewsReader Wikinews corpus (van Erp *et al.* (2015)). The Wikinews corpus consists of 120 English news articles on 4 topics: Airbus, Apple, automotive industry (GM, Chrysler and Ford), stock market news. Each topic has 30 articles. We translated the 120 English articles to Spanish, Dutch and Italian and texts in all languages have been annotated according to the NewsReader annotation scheme. We processed the English, Spanish and Dutch Wikinews articles through the respective pipelines, as described in Agerri *et al.* (2015). After generating the NAF files, we applied the NAF2SEM process to each language data set. We made a small modification to the original algorithm that compares events on the basis of their lemmas. To be able to compare events across languages, we used the ILI-concept references of the predicates to represent events. Spanish and Dutch events can thus be mapped to English events provided they were linked to the same concept.[20] We generated RDF-TRiG files for the 4 Wikinews corpora for English, Spanish and Dutch. We implemented a triple counter to compare the data created for each language data set. The triple counter generates statistics for the following information:

[20]This also means that we lump together event-instances that are normally kept separate because they do not share the same time and participant. The comparison therefore does not tell us anything about the cross-lingual event-coreference.

```
<entity id="e1" type="MISC">
   <references>
      <!--Boeing 787-->
     <span><target id="t8"/><target id="t9"/></span>
   </references>
   <externalReferences>
     <externalRef confidence="1.0" reference="http://DBpedia.org/resource/Boeing_787_Dreamliner"
     reftype="en" resource="spotlight_v1""/>
     </externalRef>
   </externalReferences>
 </entity>

 <predicate id="pr6">
   <!--purchase-->
   <externalReferences>
     <externalRef reference="purchase.01" resource="PropBank"/>
     <externalRef reference="obtain-13.5.2" resource="VerbNet"/>
     <externalRef reference="obtain-13.5.2-1" resource="VerbNet"/>
     <externalRef reference="Commerce_buy" resource="FrameNet"/>
     <externalRef reference="purchase.01" resource="PropBank"/>
     <externalRef reference="Buying" resource="ESO"/>
     <externalRef reference="contextual" resource="EventType"/>
     <externalRef reference="ili-30-02207206-v" resource="WordNet"/>
   </externalReferences>
   <span><target id="t28"/></span>
   <role id="rl9" semRole="A0">
     <!--Officials from the People 's Republic of China-->
     <externalReferences>
       <externalRef reference="obtain-13.5.2@Agent" resource="VerbNet"/>
       <externalRef reference="Commerce_buy@Buyer" resource="FrameNet"/>
       <externalRef reference="purchase.01@0" resource="PropBank"/>
       <externalRef reference="Buying@possession-owner_2" resource="ESO"/>
     </externalReferences>
     <span> <target head="yes" id="t17"/>...<target id="t24"/></span>
   </role>
   <role id="rl10" semRole="A1">
     <!--60 Boeing 787 Dreamliner aircraft-->
     <externalReferences>
       <externalRef reference="obtain-13.5.2@Theme" resource="VerbNet"/>
       <externalRef reference="Commerce_buy@Goods" resource="FrameNet"/>
       <externalRef reference="purchase.01@1" resource="PropBank"/>
     </externalReferences>
     <span><target id="t29"/>... <target head="yes" id="t33"/></span>
   </role>
   <role id="rl11" semRole="AM-LOC">
     <!--in a deal worth US$ 7.2 bn-->
     <span>
       <target head="yes" id="t34"/>...<target id="t40"/>
     </span>
   </role>
 </predicate>

 <timex3 id="tmx5" type="DATE" value="2005-01-29">
   <!--today-->
   <span><target id="w183"/> </span>
 </timex3>
```

Figure 11: Example of representation of entities, events and roles from an English Wikinews fragment

```
<entity id="e1" type="MISC">
      <references>
        <span>
          <!--Boeing 787-->
          <target id="t8"/>
          <target id="t9"/>
        </span>
      </references>
      <externalReferences>
        <externalRef confidence="1.0" reference="http://es.DBpedia.org/resource/Boeing_787"
        reftype="es" resource="spotlight_v1">
          <externalRef confidence="1.0" reference="http://DBpedia.org/resource/Boeing_787_Dreamliner"
          reftype="en" resource="wikipedia-db-esEn"/>
        </externalRef>
      </externalReferences>
    </entity>

  <predicate id="pr3">
      <!--comprar-->
      <externalReferences>
        <externalRef reference="comprar.1.benefactive" resource="AnCora"/>
        <externalRef reference="get-13.5.1" resource="VerbNet"/>
        <externalRef reference="obtain-13.5.2" resource="VerbNet"/>
        <externalRef reference="obtain-13.5.2-1" resource="VerbNet"/>
        <externalRef reference="Commerce_buy" resource="FrameNet"/>
        <externalRef reference="buy.01" resource="PropBank"/>
        <externalRef reference="purchase.01" resource="PropBank"/>
        <externalRef reference="Buying" resource="ESO"/>
        <externalRef reference="contextual" resource="EventType"/>
        <externalRef reference="ili-30-02207206-v" resource="WordNet"/>
        <externalRef reference="ili-30-02646757-v" resource="WordNet"/>
      </externalReferences>
      <span>
        <target id="t34"/>
      </span>
      <role id="rl5" semRole="arg1">
        <!--60 aviones Boeing 787 Dreamliner-->
        <externalReferences>
          <externalRef reference="get-13.5.1@Theme" resource="VerbNet"/>
          <externalRef reference="obtain-13.5.2@Theme" resource="VerbNet"/>
          <externalRef reference="Commerce_buy@Goods" resource="FrameNet"/>
          <externalRef reference="buy.01@1" resource="PropBank"/>
          <externalRef reference="purchase.01@1" resource="PropBank"/>
        </externalReferences>
        <span>
          <target id="t35"/>
          <target head="yes" id="t36"/>...<target id="t39"/>
        </span>
      </role>
      <role id="rl6" semRole="argM">
        <!--en un acuerdo por valor de 7.200 millones de US$-->
        <span>
          <target head="yes" id="t40"/>...<target id="t49"/>
        </span>
      </role>
    </predicate>

    <timex3 id="tx2" type="DATE" value="2005-01-29">
      <!--29 de enero del 2005-->
      <span>
        <target id="w20"/>... <target id="w24"/>
      </span>
    </timex3>
```

Figure 12: Example of representation of entities, events and roles from a Spanish Wikinews fragment

```
<entity id="e2" type="MISC">
   <references>
     <span>
       <!--Airbus A320-->
       <target id="t_13"/>
       <target id="t_14"/>
     </span>
   </references>
   <externalReferences>
     <externalRef confidence="1.0" reference="http://nl.DBpedia.org/resource/Airbus_A320"
     reftype="nl" resource="spotlight_v1">
       <externalRef confidence="1.0" reference="http://DBpedia.org/resource/Airbus_A320_family"
       reftype="en" resource="wikipedia-db-nlEn"/>
     </externalRef>
   </externalReferences>
 </entity>

     <predicate id="pr6">
   <!--kopen-->
   <externalReferences>
     <externalRef reference="r_v-4101" resource="Cornetto"/>
     <externalRef reference="ili-30-02207206-v" resource="WordNet"/>
   </externalReferences>
   <span>
     <target id="t_17"/>
   </span>
   <role id="r8" semRole="Arg1">
     <!--twintig Airbus A320 passagiersvliegtuigen-->
     <span>
       <target id="t_12"/>... <target head="yes" id="t_15"/>
     </span>
   </role>
   <role id="r10" semRole="ArgM-PNC">
     <!--voor een-->
     <span>
       <target head="yes" id="t_18"/>
       <target id="t_19"/>
     </span>
   </role>
 </predicate>

 <timex3 id="tmx3" type="DATE" value="2009-06-18">
   <!--donderdag-->
   <span>
     <target id="w5"/>
   </span>
 </timex3>
```

Figure 13: Example of representation of entities, events and roles from a Dutch Wikinews fragment

1. Entities with a DBpedia URI with the number of mentions
2. Entities without a DBpedia URI with the number of mentions
3. Events represented by their ILI-reference with the number of mentions
4. Frequency count of the roles relating events and entities
5. Frequency counts of the triples relating events with ILI-references and entities through these roles

We compared the Spanish and Dutch extracted data against the English data, calculating the coverage of the English mentions by the other languages for the above data. Note that we cannot calculate recall through this message. If the software detects a DBpedia entity E in English in sentence A and not in the translated Spanish sentence T(A) but the same entity E is detected in another Spanish sentence T(B) while it is not detected in the English sentence B, then our current counts give 1 mention in English and 1 mention in Spanish and 100% coverage. This may still count as coverage but not as recall because we do not know whether the Spanish mention in T(B) corresponds with the English mention in A.[21] The coverage scores we give below are thus just a rough approximation. A true comparison requires a cross-lingual annotation that also aligns the mentions of entities and events with respect to the same instances. Furthermore, we only considered the data for events that are considered as contextual according to the software. Source introducing events (e.g. speech acts) and grammatical events (to be, to start, to begin) are excluded for the time being.

The NAF2SEM module is ignorant of the language of the NAF file. This means that we can process English, Spanish and Dutch NAF files as if they are different sources, just as we process multiple English NAF files. The module will merge all entities and events according to the implemented heuristics and generate a single RDF-TRiG for all the sources. This means that we can process all the English NAF files with all the NAF file for the translations and evaluate to what extend the extracted information is identical. In the ideal case, the same data should be extracted across the languages. Therefore, merging the NAF files across languages should result in exactly the same numbers of entities, events and triples. Merging the data set can be seen as an extreme test for the cross-lingual extraction and compatibility of the different NLP-pipelines. Not all extracted information can be compared. Events that are not mapped to WordNet concepts (ILI-records) are represented by their label, which is different across the languages. The same folds for triples with these events. Entities that are not mapped to DBpedia are compared. It may happen that the name is the same across languages.

In Figure 14, we show the RDF-TRiG result of merging English, Spanish and Dutch NAF files for one entity and two events. The entity Airbus was found in NAF files for all 3 languages with 6 mentions in the English source, 7 mentions in the Spanish source and 4

[21] In some case a single English sentence has been translated in more than one sentence in another language

```
<http://DBpedia.org/resource/Airbus>
            rdfs:label      "Airbus" , "Airbus ," ;
            gaf:denotedBy
            <http://www.newsreader-project.eu/data/cars/1816_Airbus_wins_Qatar_Airways_order_worth_15bn_dutch_utf8.nl#char=31,37> ,
            <http://www.newsreader-project.eu/data/cars/1816_Airbus_wins_Qatar_Airways_order_worth_15bn_dutch_utf8.nl#char=564,570> ,
            <http://www.newsreader-project.eu/data/cars/1816_Airbus_wins_Qatar_Airways_order_worth_15bn_dutch_utf8.nl#char=655,661> ,
            <http://www.newsreader-project.eu/data/cars/1816_Airbus_wins_Qatar_Airways_order_worth_15bn_dutch_utf8.nl#char=911,917> ,
            <http://www.newsreader-project.eu/data/cars/../Annotation-Spanish/Boeing/Boeing_txt/1816_Airbus_wins.txt.es#char=0,6> ,
            <http://www.newsreader-project.eu/data/cars/../Annotation-Spanish/Boeing/Boeing_txt/1816_Airbus_wins.txt.es#char=615,621> ,
            <http://www.newsreader-project.eu/data/cars/../Annotation-Spanish/Boeing/Boeing_txt/1816_Airbus_wins.txt.es#char=716,722> ,
            <http://www.newsreader-project.eu/data/cars/../Annotation-Spanish/Boeing/Boeing_txt/1816_Airbus_wins.txt.es#char=945,951> ,
            <http://www.newsreader-project.eu/data/cars/../Annotation-Spanish/Boeing/Boeing_txt/1816_Airbus_wins.txt.es#char=450,456> ,
            <http://www.newsreader-project.eu/data/cars/../Annotation-Spanish/Boeing/Boeing_txt/1816_Airbus_wins.txt.es#char=381,387> ,
            <http://www.newsreader-project.eu/data/cars/../Annotation-Spanish/Boeing/Boeing_txt/1816_Airbus_wins.txt.es#char=135,141> ,
            <http://en.wikinews.org/wiki/Airbus_wins_Qatar_Airways_order_worth_$15bn.en#char=1,7> ,
             <http://en.wikinews.org/wiki/Airbus_wins_Qatar_Airways_order_worth_$15bn.en#char=93,100> ,
            <http://en.wikinews.org/wiki/Airbus_wins_Qatar_Airways_order_worth_$15bn.en#char=356,362> ,
            <http://en.wikinews.org/wiki/Airbus_wins_Qatar_Airways_order_worth_$15bn.en#char=641,647> ,
            <http://en.wikinews.org/wiki/Airbus_wins_Qatar_Airways_order_worth_$15bn.en#char=153,156> ,
            <http://en.wikinews.org/wiki/Airbus_wins_Qatar_Airways_order_worth_$15bn.en#char=872,878> .


nwrontology:ili-30-01451842-v-and-ili-30-01847845-v-and-ili-30-01840238-v-and-ili-30-02140965-v
            a               sem:Event , eso:Motion , eso:Transportation , eso:Translocation ,
            fn:Bringing , fn:Motion , fn:Operate_vehicle , fn:Ride_vehicle , fn:Self_motion ,
            <http://www.newsreader-project.eu/ontologies/wordnet3.0/ili-30-01451842-v> ,
            <http://www.newsreader-project.eu/ontologies/wordnet3.0/ili-30-01847845-v> ,
            <http://www.newsreader-project.eu/ontologies/wordnet3.0/ili-30-01840238-v> ,
            <http://www.newsreader-project.eu/ontologies/wordnet3.0/ili-30-02140965-v> ,
            nwrontology:volar.1.default ,
            nwrontology:r_v-9615 ;
            rdfs:label      "vliegen" , "fly" , "volar" ;
            gaf:denotedBy  <http://en.wikinews.org/wiki/Airbus_wins_Qatar_Airways_order_worth_$15bn.en#char=595,598> ,
            <http://www.newsreader-project.eu/data/cars/../Annotation-Spanish/Boeing/Boeing_txt/1816_Airbus_wins.txt.es#char=639,644> ,
            <http://www.newsreader-project.eu/data/cars/1816_Airbus_wins_Qatar_Airways_order_worth_15bn_dutch_utf8.nl#char=626,633> .


nwrontology:ili-30-00156601-v-and-ili-30-00157844-v-and-ili-30-00240293-v-and-ili-30-00154778-v-and-ili-30-00153263-v
            a               sem:Event ,
            fn:Change_position_on_a_scale , fn:Cause_expansion , fn:Expansion , eso:QuantityChange ,
            <http://www.newsreader-project.eu/ontologies/wordnet3.0/ili-30-00156601-v> ,
            <http://www.newsreader-project.eu/ontologies/wordnet3.0/ili-30-00157844-v> ,
            <http://www.newsreader-project.eu/ontologies/wordnet3.0/ili-30-00240293-v> ,
            <http://www.newsreader-project.eu/ontologies/wordnet3.0/ili-30-00154778-v> ,
            <http://www.newsreader-project.eu/ontologies/wordnet3.0/ili-30-00153263-v> ,
            nwrontology:r_v-9010 ;
            nwrontology:aumentar.1.benefactive ,
            rdfs:label      "verhogen" , "aumentar" ;
            gaf:denotedBy  <http://www.newsreader-project.eu/data/cars/../Annotation-Spanish/Boeing/Boeing_txt/1815_Ryanair_exercises.txt.es#char=1093,1100> ,
            <http://www.newsreader-project.eu/data/cars/.nl#char=1158,1166> ,
            <http://www.newsreader-project.eu/data/cars/1815_Ryanair_exercises_options_on_five_Boeing_737s_dutch_utf8.nl#char=1158,1166> .
```


Figure 14: RDF-TRiG representation of entities and events merged from English, Spanish and Dutch Wikinews

mentions in the Dutch source.[22] Similarly, the event labeled as “vliegen” , “fly” , “volar” has been matched through the ILI-concepts and now has a mention in each language. Finally, we show an event that has been matched across Spanish and Dutch but not with any English event. Such a match does not count for the coverage measure against English.

In the next subsections, we show the statistics for the 4 corpora and 3 languages. We also provide some statistics for the merging of data against the English results. All figures indicate coverage.

5.1 Crosslingual extraction of entities

In Table 13, we give the totals of DBpedia entities extracted for English and the other languages: Spanish and Dutch. These totals (column header *Nr* represent mentions in each subcorpus. The column *Values* gives the total number of unique DBpedia entities extracted for the English Wikenews files. For each language, Spanish and Dutch, we give the macro and micro coverage (C). The English figures show that there is some variation across the 4 corpora. The *stock_market* corpus contains about the same amount of entities as the *apple* corpus (76 against 74) but the *apple* corpus has 3 times more mentions. The figures for Spanish and Dutch only refer to the mentions of entities that have values for English. So if in Spanish or Dutch many DBpedia entities are extracted that are not extracted in English then these are not considered. Inspection showed that this actually not the case. For Spanish, we see that the coverage is around 28%, except for the *gm_chrysler_ford* corpus which score less than 17%. We see that Dutch has only half these results except for the *gm_chrysler_ford* corpus, where it scores just below 13%. The difference between micro and macro average for the *gm_chrysler_ford* corpus in Spanish seems to suggest that the performance for the most frequent entities has an impact. Dutch performing less is something we see for all the results below. This is not so surprising since the English and Spanish pipeline have been built with more or less the same software by the same group. Also, the Spanish wordnet is built through the expand method and therefore has a very direct relation with English WordNet. The Dutch pipeline has been built mostly with completely different software and the Dutch wordnet is built using the merge approach (a different semantics and more indirect relation with English, also a lower coverage). Differences in the output do not mean that the Dutch interpretation of the text is wrong. We are just comparing it with the English automatic interpretation.

Overall, the coverage is a bit low both for Spanish and English. This is however the first time we have been able to do such a comparison. We can now analyse the differences for the most frequent entities to find ways to improve the software. The Tables 14, 15, 16, 17 show the top 15 entities most frequent in English with the number in Spanish and Dutch for all 4 corpora. For each entity, we show the number of mentions and the proportion of English mentions covered. If there are more mentions in Spanish or Dutch, the coverage is 100%. There are a few interesting observations to be made. An entity such as *France*

[22]We extended the source file names in the denotedBy relation with a language tag to keep them apart. “.nl” for Dutch, “.es” for Spanish and “.en” for English

has 57 mentions in the Airbus corpus for English, only 1 for Spanish and 0 for Dutch. We checked the Dutch output. *France* is detected correctly but it has not been mapped to an event as a participant and therefore was not included in the RDF-TRiG output. *United_States_dollar* with 39 mentions in English, on the other hand, turned out to be an error in the English pipeline that is not mirrored by the other languages. The English pipeline erroneously linked mentions of the US to the dollar instead of the country. Further analysis will be used to find systematic errors by the pipelines, which will make the results more interoperable.

Table 13: DBpedia entities extracted for English, Spanish and Dutch Wikinews with proportion of coverage, measured as macro and micro coverage

	English		Spanish			Dutch		
DBpedia entities	**Values**	**Nr**	**Nr**	**macro C**	**micro C**	**Nr**	**macro C**	**micro C**
airbus	96	1000	386	28.66	38.60	189	14.42	18.90
apple	74	1508	508	32.22	33.69	230	12.39	15.25
gm_chysler_ford	65	1194	332	17.78	27.81	241	12.65	20.18
stock_market	76	465	127	25.58	27.31	55	8.18	11.83

Table 14: DBpedia entities in the Wikinews Airbus corpus most frequent in English with Spanish and Dutch frequencies

DBpedia Entities	**English**	**Spanish**		**Dutch**	
Boeing	171	152	88.89	73	42.69
Airbus	123	73	59.35	34	27.64
France	57	1	1.75	0	0.00
European_Union	57	17	29.82	14	24.56
Boeing_Commercial_Airplanes	55	0	0.00	4	7.27
United_States_Air_Force	46	21	45.65	7	15.22
United_States_dollar	39	0	0.00	0	0.00
Indonesia	38	13	34.21	0	0.00
Government_Accountability_Office	36	0	0.00	0	0.00
Aer_Lingus	33	16	48.48	8	24.24
Boeing_747	30	0	0.00	0	0.00
Airbus_A320_family	22	3	13.64	1	4.55
Toulouse	16	0	0.00	1	6.25
Singapore	16	0	0.00	0	0.00
United_States_Armed_Forces	15	0	0.00	0	0.00

As explained above, we also detect entities that are not found in DBpedia. In many cases these are names of persons and organizations that are simply not listed in DBPedia but that can be identified as entities through their form. Table 18 shows the number of entities in the English corpora and the matching mentions across the other languages. We can see that in this data set about 25% of the entities is not matched to DBpedia in English. We can also see that occasionally there is still a match from Spanish and Dutch to English, although overall the coverage is low. An except is the Apple corpus for which a more substantial number of entities was matched.

Table 15: DBpedia entities in the Wikinews Apple corpus most frequent in English with Spanish and Dutch frequencies

DBpedia Entities	English	Spanish		Dutch	
Apple_Inc.	992	417	42.04	185	18.65
Steve_Waugh	131	0	0.00	0	0.00
Steve_Jobs	110	6	5.45	13	11.82
United_States_dollar	36	0	0.00	0	0.00
Cisco_Systems	20	16	80.00	2	10.00
Mac_OS_X_Tiger	19	0	0.00	0	0.00
Intel	15	7	46.67	0	0.00
MacBook_Air	14	0	0.00	0	0.00
The_Beatles	11	6	54.55	5	45.45
ITunes_Store	9	2	22.22	0	0.00
MacBook	9	7	77.78	0	0.00
United_Kingdom	7	4	57.14	0	0.00
IBM	7	0	0.00	3	42.86
Apple_Corps	6	33	100.00	22	100.00
OS_X	6	2	33.33	14	100.00

Table 16: DBpedia entities in the Wikinews GM, Chrysler, Ford corpus most frequent in English with Spanish and Dutch frequencies

DBpedia Entities	English	Spanish		Dutch	
General_Motors	296	96	32.43	131	44.26
Ford_Motor_Company	171	87	50.88	29	16.96
Chrysler	136	33	24.26	5	3.68
United_States_dollar	108	0	0.00	2	1.85
Canada	78	0	0.00	9	11.54
United_States	37	29	78.38	35	94.59
Daimler_AG	36	5	13.89	4	11.11
Fiat	32	28	87.50	0	0.00
Ford_Taurus	32	0	0.00	1	3.13
Federal_government_of_the_United_States	23	0	0.00	0	0.00
United_Auto_Workers	23	10	43.48	0	0.00
Barack_Obama	20	4	20.00	2	10.00
Romania	18	3	16.67	2	11.11
Ford_Motor_Company_of_Australia	15	0	0.00	0	0.00
Ontario	12	2	16.67	0	0.00

Table 17: DBpedia entities in the Wikinews stock market corpus most frequent in English with Spanish and Dutch frequencies

DBpedia Entities	English	Spanish		Dutch	
United_States_dollar	77	0	0.00	0	0.00
Dow_Jones_Industrial_Average	57	20	35.09	4	7.02
Russia	52	0	0.00	3	5.77
United_States	27	47	100.00	28	100.00
Japan	22	13	59.09	3	13.64
Federal_Reserve_System	22	3	13.64	3	13.64
American_International_Group	18	13	72.22	0	0.00
NASDAQ	16	15	93.75	0	0.00
United_Kingdom	13	1	7.69	2	15.38
FTSE_100_Index	12	2	16.67	1	8.33
European_Union	11	1	9.09	1	9.09
Nikkei_225	8	5	62.50	0	0.00
China	8	0	0.00	0	0.00
Economy_of_Japan	7	0	0.00	0	0.00
Bank_of_America	7	1	14.29	1	14.29

Table 18: NON-DBpedia entities extracted for English, Spanish and Dutch Wikinews with proportion of coverage, measured as macro and micro coverage

	English		Spanish			Dutch		
NON-DBP entities	**Values**	**Nr**	**Nr**	**macro C**	**micro C**	**Nr**	**macro C**	**micro C**
airbus	23	27	0	0.00	0.00	1	2.17	3.70
apple	19	80	57	10.16	71.25	17	6.63	21.25
gm_chysler_ford	17	49	5	7.82	10.20	5	7.65	10.20
stock_market	30	55	2	6.67	3.64	4	10.47	7.27

The NON-DBpedia entities most frequent in English (Airbus corpus) are: *EADS-led (3), AirTanker_Ltd. (2), and R-AZ (2)*. Entities only detected in Spanish are: *Finance_de_Kuwait*, *European_Aeronautic_Defence_%26_Space_NV*, *Indonesia_Susilo_Bambang_Yudhoyono*, *Sikorky*, *Las_Fuerzas_A%E2%88%9A%C2%A9reas*, *El_A350_XWB*. Only Dutch entities are: *Luchtvaartanalist_Richard_Aboulafia*, *Boeing_35*, *Lockheed_en_Sikorsky*, *Tom_Enders*. One entity was detected in Spanish and Dutch but not in English: *Aer_Lingus_Group_PLC*.

5.2 Crosslingual extraction of events

As explained above, we represent events through the ILI-concepts that are associated with their lemmas. This approximates the representation of the concept. Furthermore in some cases, more than one concept is assigned to a single lemma. To compare such lists of concepts, we checked if there was at least one intersecting ILI-concept across events to decide on a match.

We can see in Table 19 that there is a broader set of values than for entities. The proportions of matched events from Spanish and Dutch to English are just a little bit lower than for DBpedia entities but not much. This is promising since matching events is more difficult than matching entities. We see again that Dutch scores lower than Spanish. We expect that this is due to the fact that the Spanish wordnet is a direct extension of the English wordnet, whereas the Dutch wordnet is built independently. As with the entities, coverage is low and there is a lot of room for improvement.

Table 19: ILI-based events extracted for English, Spanish and Dutch Wikinews with proportion of coverage, measured as macro and micro coverage

	English		Spanish			Dutch		
ILI Events	**Values**	**Nr**	**Nr**	**macro C**	**micro C**	**Nr**	**macro C**	**micro C**
airbus	155	507	168	30.16	33.14	98	13.95	19.33
apple	152	426	137	27.03	32.16	59	17.42	13.85
gm_chysler_ford	154	579	116	17.16	20.03	57	11.83	9.84
stock_market	130	490	103	24.71	21.02	35	12.39	7.14

In the Tables 20, 21, 22, 23 we show the ILI-based events that are most frequent in English for the 4 corpora. If there is more than one ILI-record assigned, we only list the synonyms for the first synset.[23] The overview in these tables can be used to improve the mapping of concepts across the languages. In the 3rd year of the project, we will manually improve the mapping for the most frequent concepts, which will have a big impact on the interoperability of the processing across the languages.

[23]In a sense key the first digit indicate the synset type, the second the lexicographer file and the third a unique sense within the lexicographer's file. For an explanation of the sense keys see `http://wordnet.princeton.edu/man/senseidx.5WN.html`

Table 20: ILI-based events in the Wikinews Airbus corpus most frequent in English with Spanish and Dutch frequencies

ILI EVENTS	English	Spanish		Dutch	
fly%2:38:05;fly%2:35:00;fly%2:39:00;fly%2:38:04	50	21	42.00	5	10.00
deliver%2:35:00	19	13	68.42	6	31.58
carry%2:35:02; transport%2:35:00	18	0	0.00	5	27.78
do%2:42:02;serve%2:42:00	14	0	0.00	0	0.00
provide%2:34:00;cater%2:34:00;ply%2:34:00;supply%2:34:00;	14	14	100.00	13	92.86
construct%2:36:01;fabricate%2:36:01;manufacture%2:36:00	13	2	15.38	0	0.00
construct%2:35:00	13	5	38.46	0	0.00
interchange%2:40:01;substitute%2:40:00;exchange%2:40:02;replace%2:40:00	11	7	63.64	0	0.00
buy%2:40:00	10	17	100.00	12	100.00
ride%2:38:00	9	0	0.00	0	0.00
delay%2:42:00	9	6	66.67	0	0.00
have%2:40:03;receive%2:40:00	9	0	0.00	6	66.67
create%2:36:03;make%2:36:01;produce%2:36:00	9	2	22.22	0	0.00
change%2:30:04;commute%2:30:02;convert%2:30:02;exchange%2:30:00	9	5	55.56	0	0.00
record%2:32:01;tape%2:32:00	8	0	0.00	0	0.00

Table 21: ILI-based events in the Wikinews Apple corpus most frequent in English with Spanish and Dutch frequencies

ILI EVENTS	English	Spanish		Dutch	
deal%2:40:00;sell%2:40:01;trade%2:40:11	34	22	64.71	0	0.00
hold%2:35:00;take_hold%2:35:00	14	0	0.00	0	0.00
build%2:36:00;construct%2:36:00;make%2:36:13	12	6	50.00	0	0.00
hive_away%2:40:00;lay_in%2:40:00;put_in%2:40:00;salt_away%2:40:00; stack_away%2:40:00;stash_away%2:40:00;store%2:40:00	11	2	18.18	0	0.00
display%2:39:00;exhibit%2:39:01;expose%2:39:00	10	24	100.00	4	40.00
control%2:35:00;operate%2:35:00	10	0	0.00	0	0.00
bring%2:35:00;convey%2:35:00;fetch%2:35:00;get%2:35:03	10	4	40.00	0	0.00
case%2:35:01;encase%2:35:00;incase%2:35:00	9	0	0.00	0	0.00
acquire%2:40:06;gain%2:40:01;win%2:40:00	8	0	0.00	0	0.00
offer_up%2:42:00;offer%2:42:00	7	17	100.00	0	0.00
appoint%2:41:01;constitute%2:41:00;name%2:41:00;nominate%2:41:00	7	0	0.00	0	0.00
go%2:38:00;locomote%2:38:00;move%2:38:03;travel%2:38:00	7	0	0.00	0	0.00
increase%2:30:00	7	3	42.86	1	14.29
drop%2:30:00	6	1	16.67	0	0.00
file_away%2:32:00;file%2:32:0	6	0	0.00	0	0.00

Table 22: ILI-based events in the Wikinews GM, Chrysler, Ford corpus most frequent in English with Spanish and Dutch frequencies

ILI EVENTS	English	Spanish		Dutch	
deal%2:40:00;sell%2:40:01;trade%2:40:11	60	10	16.67	0	0.00
create%2:36:03;make%2:36:01;produce%2:36:00	23	4	17.39	0	0.00
construct%2:36:01;fabricate%2:36:01;manufacture%2:36:00	19	7	36.84	0	0.00
close%2:35:00;shut%2:35:00	18	12	66.67	5	27.78
lend%2:40:00;loan%2:40:00	16	2	12.50	0	0.00
acquire%2:40:06;gain%2:40:01;win%2:40:00	16	0	0.00	3	18.75
aid%2:41:00;assist%2:41:02;help%2:41:00	16	2	12.50	2	12.50
have%2:40:03;receive%2:40:00	14	0	0.00	3	21.43
decease%2:30:00;pass_away%2:30:00	13	0	0.00	0	0.00
decline%2:30:01;worsen%2:30:00	11	0	0.00	0	0.00
offer_up%2:42:00;offer%2:42:00	11	6	54.55	0	0.00
increase%2:30:00	11	10	90.91	2	18.18
mail%2:32:00;post%2:32:02;send%2:32:00	10	0	0.00	0	0.00
pay%2:40:00	9	5	55.56	18	100.00
deal%2:41:09	9	1	11.11	0	0.00

Table 23: ILI-based events in the Wikinews stock market corpus most frequent in English with Spanish and Dutch frequencies

ILI EVENTS	English	Spanish		Dutch	
decrease%2:30:00;diminish%2:30:00;fall%2:30:06;lessen%2:30:00	73	1	1.37	0	0.00
deal%2:40:00;sell%2:40:01;trade%2:40:11	34	4	11.76	0	0.00
drop%2:30:00	29	11	37.93	0	0.00
even_out%2:35:00;even%2:35:00;flush%2:35:03;level%2:35:01	26	0	0.00	0	0.00
increase%2:30:00	19	5	26.32	1	5.26
climb_up%2:38:00;climb%2:38:00;go_up%2:38:01;mount%2:38:00	16	1	6.25	6	37.50
lend%2:40:00;loan%2:40:00	14	3	21.43	0	0.00
give%2:40:06;turn_over%2:40:00;hand%2:40:00;pass_on%2:40:01;pass%2:40:00;reach%2:40:00	12	9	75.00	1	8.33
swap%2:40:00;swop%2:40:00;switch%2:40:00;trade%2:40:02	12	1	8.33	0	0.00
farm%2:36:00;grow%2:36:00;produce%2:36:05;raise%2:36:03	10	3	30.00	0	0.00
bring%2:35:00;convey%2:35:00;fetch%2:35:00;get%2:35:03	10	5	50.00	0	0.00
record%2:32:01;tape%2:32:00	8	12	100.00	0	0.00
create%2:36:03;make%2:36:01;produce%2:36:00	7	1	14.29	1	14.29
come_up%2:38:02;come%2:38:00	6	0	0.00	0	0.00
break_down%2:38:00;collapse%2:38:02;crumple%2:38:00;tumble%2:38:03;crumble%2:38:00	6	2	33.33	0	0.00

5.3 Crosslingual extraction of relations

Finally, we compared the actual triples extracted across the languages. The triples represent the actual statements, where we only consider triples where the ILI-based event is the subject. In Table 24 we show the comparison of the predicates within the triples, representing the role relations between events and participants. Both Spanish and Dutch perform lower than English but the results are reasonable for a difficult task as semantic-role-labeling. Spanish performance is a little bit higher than Dutch but the difference is small, especially compared with the previous results.

Table 24: Triple predicates extracted for English, Spanish and Dutch Wikinews with proportion of coverage, measured as macro and micro coverage

	English		Spanish			Dutch		
PREDICATES	**Values**	**Nr**	**Nr**	**macro C**	**micro C**	**Nr**	**macro C**	**micro C**
airbus	75	2302	901	29.53	39.14	770	15.57	33.45
apple	74	2440	1099	42.51	45.04	806	11.96	33.03
gm_chysler_ford	74	2490	668	29.89	26.83	793	13.42	31.85
stock_market	62	2381	1022	24.56	42.92	690	10.79	28.98

Tables 25, 26, 27, 28 give the predicates most frequent in the English data. The most frequent predicates are *hasAcor* and the PropBank roles such as *A0*, *A1*, *A2*. The FrameNet Element predicates are less frequent. Remarkably, the ESO predicates are frequent in English and Spanish. These have not yet been added to the Dutch pipeline and therefore cannot result in any scores for Dutch.

Finally, Table 29 shows the coverage results for the actual triples that relate ILI-based events with entities through the above predicates. Although many triples are generated for English, the coverage by Spanish and Dutch are very low. The lowest performance for Dutch is partially explained by the fact that the ILI-events in Dutch are based in a single sense, whereas the English and Spanish events are represented by a series of senses. Furthermore, having a full triple match across the language is a challenge and there is a lot of room for improvement by first addressing the event and entity interoperability.

Since triples usually are mentioned only once and at most a few times in the corpora

Table 25: Triple predicates that are most frequent in the English Airbus Wikinews corpus with coverage in Spanish and Dutch

PREDICATES	English	Spanish		Dutch	
hasActor	341	156	45.75	154	45.16
A0	178	64	35.96	76	42.70
A1	92	59	64.13	46	50.00
A2	58	29	50.00	1	1.72
AM-LOC	39	0	0.00	1	2.56
hasPlace	39	0	0.00	1	2.56
eso#translocation-theme	14	4	28.57	0	0.00
eso#possession-owner_2	9	8	88.89	0	0.00
eso#translocation-source	9	3	33.33	0	0.00
A3	8	4	50.00	1	12.50
eso#possession-theme	8	8	100.00	0	0.00
eso#possession-owner_1	8	3	37.50	0	0.00
Manufacturer	7	1	14.29	0	0.00
Theme	6	3	50.00	5	83.33
A4	5	0	0.00	0	0.00

Table 26: Triple predicates that are most frequent in the English Apple Wikinews corpus with coverage in Spanish and Dutch

PREDICATES	English	Spanish		Dutch	
hasActor	349	208	59.60	153.00	43.84
A0	192	100	52.08	81.00	42.19
A1	115	86	74.78	52.00	45.22
AM-LOC	21	0	0.00	1.00	4.76
hasPlace	21	0	0.00	1.00	4.76
eso#possession-owner_1	19	6	31.58	0.00	0.00
Seller	13	3	23.08	0.00	0.00
A3	12	0	0.00	0.00	0.00
eso#translocation-theme	11	4	36.36	0.00	0.00
Creator	7	5	71.43	0.00	0.00
Attribute	6	5	83.33	0.00	0.00
eso#quantity-theme	6	5	83.33	0.00	0.00
eso#possession-theme	6	4	66.67	0.00	0.00
eso#possession-owner_2	6	2	33.33	0.00	0.00
Item	5	5	100.00	1.00	20.00

Table 27: Triple predicates that are most frequent in the English GM, Chrysler, Ford Wikinews corpus with coverage in Spanish and Dutch

PREDICATES	**English**	**Spanish**		**Dutch**	
hasActor	355	124	34.93	158	44.51
A0	168	46	27.38	93	55.36
A1	123	59	47.97	42	34.15
AM-LOC	48	0	0.00	1	2.08
hasPlace	48	0	0.00	1	2.08
eso#possession-owner_1	22	3	13.64	0	0.00
eso#quantity-theme	14	2	14.29	0	0.00
Item	13	1	7.69	0	0.00
eso#possession-owner_2	13	5	38.46	0	0.00
Seller	11	2	18.18	0	0.00
Manufacturer	9	3	33.33	0	0.00
Creator	8	0	0.00	0	0.00
eso#translocation-theme	7	0	0.00	0	0.00
Cook	6	0	0.00	0	0.00
Road	6	0	0.00	0	0.00

Table 28: Triple predicates that are most frequent in the English stock market Wikinews corpus with coverage in Spanish and Dutch

PREDICATES	**English**	**Spanish**		**Dutch**	
hasActor	297	150	50.51	146	49.16
A1	139	75	53.96	36	25.90
A0	125	64	51.20	83	66.40
eso#quantity-theme	58	12	20.69	0	0.00
Item	57	1	1.75	1	1.75
AM-LOC	32	0	0.00	0	0.00
hasPlace	32	0	0.00	0	0.00
A2	30	11	36.67	5	16.67
Road	17	11	64.71	0	0.00
Attribute	12	5	41.67	1	8.33
Message	11	10	90.91	0	0.00
Source	6	11	100.00	0	0.00
eso#possession-owner_2	6	0	0.00	0	0.00
eso#possession-owner_1	6	1	16.67	0	0.00
Goal	6	11	100.00	1	16.67

Table 29: ILI-based Triples extracted for English, Spanish and Dutch Wikinews with proportion of coverage, measured as macro and micro coverage

	English		**Spanish**			**Dutch**		
TRIPLES	**Values**	**Nr**	**Nr**	**macro C**	**micro C**	**Nr**	**macro C**	**micro C**
airbus	619	632	34	5.33	5.38	24	3.88	3.80
apple	491	541	81	12.25	14.97	5	1.02	0.92
gm_chysler_ford	686	765	50	7.22	6.54	5	0.51	0.65
stock_market	575	665	26	4.17	3.91	2	0.35	0.30

(30 articles only), it makes no sense to show frequency tables of triples.

```
Triples in Spanish and Dutch but not in English
ili-30-02140033-v[display\%2:39:00;exhibit\%2:39:01;expose\%2:39:00;]:A0:Apple\_Inc.
ili-30-01323958-v[kill\%2:35:00;]:A1:Dick\_Cheney
ili-30-02207206-v[buy\%2:40:00;purchase\%2:40:00;]:A1:European\_Union

Triples in all 3 languages
ili-30-02251743-v[pay\%2:40:00;]:Buyer:Apple\_Inc.
ili-30-02207206-v[buy\%2:40:00;purchase\%2:40:00;]:hasActor:European\_Union
ili-30-01451842-v[fly\%2:35:00;]:hasActor:Airbus\_A380
ili-30-01451842-v[fly\%2:35:00;]:Vehicle:Airbus\_A380
ili-30-01451842-v[fly\%2:35:00;]:Driver:Airbus\_A380
ili-30-01451842-v[fly\%2:35:00;]:Theme:Airbus\_A380
ili-30-01451842-v[fly\%2:35:00;]:Source:Airbus\_A380
ili-30-01451842-v[fly\%2:35:00;]:Agent:Airbus\_A380
ili-30-01451842-v[fly\%2:35:00;]:A0:Airbus\_A380
ili-30-02208537-v[charter\%2:40:00;engage\%2:40:00;hire\%2:40:00;
lease\%2:40:00;rent\%2:40:00;take\%2:40:03;]:hasActor:Boeing
ili-30-02208537-v[charter\%2:40:00;engage\%2:40:00;hire\%2:40:00;
lease\%2:40:00;rent\%2:40:00;take\%2:40:03;]:A0:Boeing
ili-30-02327200-v[furnish\%2:40:00;render\%2:40:02;
provide\%2:40:00;supply\%2:40:00;]:Theme:Airbus
ili-30-02207206-v[buy\%2:40:00;purchase\%2:40:00;]:A0:Airbus
ili-30-02207206-v[buy\%2:40:00;purchase\%2:40:00;]:hasActor:Airbus
ili-30-02327200-v[furnish\%2:40:00;render\%2:40:02;
provide\%2:40:00;supply\%2:40:00;]:hasActor:Airbus
```

Figure 15: Identical triples across different languages

5.4 Conclusions

We described the results of the first cross-lingual semantic processing of text. To our knowledge, there is no existing system that can perform such a task. Being able to merge the interpretation of text across language is a big achievement and it shows the interoperability of the NewsReader system. We have also seen that for most data types coverage is still low. The current data however provides an excellent basis for improving the interoperability. We have also seen that differences in implementation have an impact on the comparability. Spanish results are more closer to English because most of the NLP modules for English and Spanish are developed by the same group, whereas the Dutch pipeline is mostly based on different software. That does not mean that the output of the Spanish software is better than the Dutch software. It only means that it is more compatible. For a qualitative evaluation, we need to use the cross-lingual annotation of the Wikinews which is being completed now. In the 3rd year of the project, we hope to provide further information on the cross-lingual quality of the different software.

6 Perspectives

In this section, we describe the design and first steps towards the implementation of a perspective and attribution module. The implementation so far is preliminary, and will be updated in the coming period. The attribution module described in 6.2 is currently restricted to factuality. Other attribution values will be derived in further updates.

6.1 Basic Perspective Module

Events are distinguished into contextual events about the domain and source events, which are speech-acts and cognitive events that relate sources to the statements about the contextual events. In NewsReader, we represent both types of events as instances, i.e. they are both events involving participants and bound by time and possibly a place. The perspective layer is generated on top of this initial event representation and is the result of combining various layers of information. Perspectives are expressed through mentions in the text but they are represented in the RDF representation on top of the instance representations of events. This means that the same perspective can be expressed in different sources, which are considered mentions of the perspective. We can also find cases where a different perspective is expressed on the same contextual statement: for example, this would be the case if one source denies a statement while another sources confirms it.

A perspective thus consists of:

- The instance representation of the source of a statement, e.g. the *CEO of Porsche Wiedeking*.
- The statement made by the source in the form of triples representing an instance of a contextual event, e.g. *Porsche increasing sales in 2004*.
- The attribution values that define the relation between the source and the contextual statement: *denial/confirmation*, *positive/negative*, *future/current*, *certain/uncertain*, etc.

According to this specification, we abstract from the way in which the source made the statement, e.g. *saying*, *shouting*, *thinking*, which is represented by the source event itself that is the basis for deriving the perspective.

To derive the perspective relations between sources and the contextual statement, we implemented a perspective module that takes several layers as input:

- source events with entities that have a roles like *prop:A0*, *fn:Statement@Speaker*, *fn:Expectation@Cognizer*, *fn:Opinion@Cognizer*.
- the contextual events that can be related to the source event through a *message* or *topic* role as defined in FrameNet or PropBank.

- the opinion layer in NAF that indicates whether the source has a positive or negative opinion with respect to some target.
- the attribution layer in NAF that indicates the future/non-future tense, the confirmation or denial and/or the certainty of the contextual event according to the source.

In order to combine these pieces of information, we need to intersect the layers. In this first version of the module, we use a very basic and simple algorithm to do this:

1. For all source events in the triples we check if it has a valid relation to an object of the proper semantic type such as *person* or *organization*. Valid relations are e.g. *pb:A0*, *fn:Statement@Speaker*, *fn:Expectation@Cognizer*, *fn:Opinion@Cognizer*. If not we ignore it.
2. We access the SRL layer in NAF to check if the event mention also has message role, e.g. *fn:Statement@Message* or *fn:Statement@Topic*. Note that the message roles are not represented in the triples so far, which is why we need to back to the NAF layer. If no such role is found we ignore the source event.
3. We take the span of message role and we check for any triples in the contextual data to see if they have mentions equal to or embedded within the span of the message. For all these triples, we create a perspective relation between the source and contextual triples.
4. We check the opinion layer in NAF to find opinions with a holder that matches the span of entities with the source role and/or an opinion expression whose span is equal to or embeds the source event span. If there is a match, we take over the opinion value (positive/negative) as a value for the perspective relation.
5. We check the attribution layer of each NAF file from which the source events and selected contextual events originate to see of there are attribution properties assigned to the event mention. If so, we copy them to the perspective relation.

The module can be adapted by defining the role constraints in the algorithm. Let us considering the following example to illustrate the algorithm:

```
Chrysler expects to sell 5,000 diesel Liberty SUVs,
President Dieter Zetsche says at a DaimlerChrysler Innovation Symposium in New York.
```

There are two perspectives expressed with respect to the *selling of 5,000 diesel Liberty SUVs*. First, the fact that this is an *expectation of Chrysler* and, secondly, that this is a *statement made by the President Dieter Zetsche*. Our text processing generates the following 3 predicate-role structures for this sentence, where we abbreviated some of the lists of span elements:

```
<!--t36 expects : A0[t35 Chrysler] A1[t37 to]-->
<predicate id="pr6">
  <!--expects-->
  <span>
    <target id="t36"/>
  </span>
  <externalReferences>
    <externalRef resource="FrameNet" reference="Expectation"/>
    <externalRef resource="FrameNet" reference="Opinion"/>
    <externalRef resource="EventType" reference="cognition"/>
  </externalReferences>
  <role id="rl16" semRole="A0">
    <!--Chrysler-->
    <span>
      <target id="t35" head="yes"/>
    </span>
    <externalReferences>
      <externalRef resource="FrameNet" reference="Expectation@Cognizer"/>
      <externalRef resource="FrameNet" reference="Opinion@Cognizer"/>
    </externalReferences>
  </role>
  <role id="rl17" semRole="A1">
    <!--to sell 5,000 diesel Liberty SUVs-->
    <span>
      <target id="t37" head="yes"/>...<target id="t42"/>
    </span>
    <externalReferences>
      <externalRef resource="FrameNet" reference="Expectation@Phenomenon"/>
      <externalRef resource="FrameNet" reference="Expectation@Topic"/>
      <externalRef resource="FrameNet" reference="Opinion@Topic"/>
    </externalReferences>
  </role>
</predicate>

  <!--t38 sell : A0[t35 Chrysler] A1[t39 5,000]-->
<predicate id="pr7">
  <!--sell-->
  <span>
    <target id="t38"/>
  </span>
  <externalReferences>
    <externalRef resource="FrameNet" reference="Commerce_sell"/>
    <externalRef resource="ESO" reference="Selling"/>
  </externalReferences>
  <role id="rl18" semRole="A0">
    <!--Chrysler-->
    <span>
      <target id="t35" head="yes"/>
    </span>
    <externalReferences>
      <externalRef resource="FrameNet" reference="Commerce_sell@Seller"/>
      <externalRef resource="ESO" reference="Selling@possession-owner_1"/>
    </externalReferences>
  </role>
  <role id="rl19" semRole="A1">
    <!--5,000 diesel Liberty SUVs-->
    <span>
      <target id="t39"/>...<target id="t42" head="yes"/>
    </span>
    <externalReferences>
      <externalRef resource="VerbNet" reference="give-13.1@Theme"/>
      <externalRef resource="FrameNet" reference="Commerce_sell@Goods"/>
      <externalRef resource="PropBank" reference="sell.01@1"/>
    </externalReferences>
  </role>
```

```
</predicate>

 <!--t47 says : A1[t35 Chrysler] A0[t44 President] AM-LOC[t48 at]-->
<predicate id="pr8">
  <!--says-->
  <span>
    <target id="t47"/>
  </span>
  <externalReferences>
     <externalRef resource="FrameNet" reference="Statement"/>
    <externalRef resource="FrameNet" reference="Text_creation"/>
  </externalReferences>
  <role id="rl20" semRole="A1">
    <!--Chrysler expects to sell 5,000 diesel Liberty SUVs-->
    <span>
      <target id="t35"/> <target id="t36" head="yes"/>...<target id="t42"/>
    </span>
    <externalReferences>
      <externalRef resource="FrameNet" reference="Statement@Message"/>
      <externalRef resource="FrameNet" reference="Statement@Topic"/>
      <externalRef resource="FrameNet" reference="Text_creation@Text"/>
      <externalRef resource="FrameNet" reference="Choosing@Chosen"/>
     </externalReferences>
  </role>
  <role id="rl21" semRole="A0">
    <!--President Dieter Zetsche-->
    <span>
      <target id="t44"/>...<target id="t46" head="yes"/>
    </span>
    <externalReferences>
      <externalRef resource="FrameNet" reference="Statement@Speaker"/>
      <externalRef resource="FrameNet" reference="Text_creation@Author"/>
      <externalRef resource="FrameNet" reference="Choosing@Cognizer"/>
     </externalReferences>
  </role>
  <role id="rl22" semRole="AM-LOC">
    <!--at a DaimlerChrysler Innovation Symposium in New York-->
    <span>
      <target id="t48" head="yes"/>...<target id="t55"/>
    </span>
  </role>
</predicate>
```

Next, the NAF2SEM modules generates two source events from this data involving the entities *Chrysler* and *Zetsche*:

```
nwr:/data/cars/2003/01/01/47VH-FG30-010D-Y3YG.xml#ev16Expect
        a sem:Event , fn:Expectation , fn:Opinion , nwrontology:SPEECH_COGNITIVE ;
        rdfs:label "expect" ;
        gaf:denotedBy nwr:/data/cars/2003/01/01/47VH-FG30-010D-Y3YG.xml#char=208,215.

dbp:resource/Chrysler
        rdfs:label          "Chrysler" , "Chrysler Group" ;
        gaf:denotedBy
        nwr:/data/cars/2003/01/01/47VH-FG30-010D-Y3YG.xml#char=36,50,
        nwr:/data/cars/2003/01/01/47VH-FG30-010D-Y3YG.xml#char=740,748 ,
        nwr:/data/cars/2003/01/01/47VH-FG30-010D-Y3YG.xml#char=199,207 ,
        nwr:/data/cars/2003/01/01/47VH-FG30-010D-Y3YG.xml#char=1114,1122> ,
        nwr:/data/cars/2003/01/01/47VH-FG30-010D-Y3YG.xml#char=130,132 .

nwr:/data/cars/entities/Liberty_SUVs
        a                   nwrontology:MISC ;
        rdfs:label           "Liberty suvs" ;
```

```
        gaf:denotedBy  nwr:/data/cars/2003/01/01/47VH-FG30-010D-Y3YG.xml#char=237,249 .


nwr:/data/cars/2003/01/01/47VH-FG30-010D-Y3YG.xml#ev16Expect
        fn:Expectation@Cognizer  dbp:resource/Chrysler;
        fn:Expectation@Phenomenon  nwr:/data/cars/entities/Liberty_SUVs ;
        fn:Expectation@Topic              nwr:/data/cars/entities/Liberty_SUVs ;
        fn:Opinion@Topic                    nwr:/data/cars/entities/Liberty_SUVs .

dbp:resource/Dieter_Zetsche
        rdfs:label          "Dieter Zetsche" ;
        gaf:denotedBy  nwr:/data/cars/2003/01/01/47VH-FG30-010D-Y3YG.xml#char=261,275 .

nwr:/data/cars/2003/01/01/47VH-FG30-010D-Y3YG.xml#ev11
        a                       sem:Event , nwrontology:SPEECH_COGNITIVE , fn:Statement;
        rdfs:label           "say" ;
        gaf:denotedBy   nwr:/data/cars/2003/01/01/47VH-FG30-010D-Y3YG.xml#char=276,280 .

nwr:/data/cars/entities/DaimlerChrysler_Innovation_Symposium
        a                      nwrontology:ORGANIZATION ;
        rdfs:label          "DaimlerChrysler Innovation Symposium" ;
        gaf:denotedBy   nwr:/data/cars/2003/01/01/47VH-FG30-010D-Y3YG.xml#char=286,322 .

nwr:/data/cars/2003/01/01/47VH-FG30-010D-Y3YG.xml#ev11Say
        fn:Statement@Speaker         dbp:resource/Dieter_Zetsche;
        sem:hasPlace  nwr:/data/cars/entities/DaimlerChrysler_Innovation_Symposium.
```

Both source events *expect* and *say* meet the first constraints that they have an entity of the proper type with a role of the type source: *fn:Expectation@Cognizer* and *fn:Statement@Speaker*. Within the set of contextual triples, we find the event *sell* and its corresponding triples:

```
nwr:/data/cars/2003/01/01/47VH-FG30-010D-Y3YG.xml#ev17Sell
          a              sem:Event , fn:Commerce_sell , eso:Selling ;
          rdfs:label     "sell" ;
          gaf:denotedBy  nwr:/data/cars/2003/01/01/47VH-FG30-010D-Y3YG.xml#char=219,223 .

 nwr:/data/cars/entities/Liberty_SUVs
          rdfs:label     "Liberty suvs" ;
          gaf:denotedBy  nwr:/data/cars/2003/01/01/47VH-FG30-010D-Y3YG.xml#char=237,249.

 nwr:/data/cars/2003/01/01/47VH-FG30-010D-Y3YG.xml#ev17Sell
          eso:possession-owner_1   dbp:resource/Chrysler ;
          fn:Commerce_sell@Seller dbp:resource/Chrysler;
          fn:Commerce_sell@Goods nwr:data/cars/entities/Liberty_SUVs .
```

Next, we intersect the mentions of the contextual triples with the role layer to see if they can be connected to the source events in the proper way. The SRL has roles for *Expectation@Topic*, *Statement@Topic* and *Statement@Message*. Their spans are defined as a list of term identifiers that need to be matched with tokens that can be matched with their offsets. In this case, we can conclude that the offsets for *sell*, *Chrysler* and *Livery_SUVs* match with these roles. Therefore, we can conclude that these triples fall within the scope of the perspective.

In a similar way, we check for opinions and attribution values to fill in further details of the perspective relation. For this example, we find the following opinion information:

```
<opinion id="o1">
      <opinion_expression polarity="positive" strength="1">
```

```
      <!--President Dieter Zetsche says at a DaimlerChrysler Innovation Symposium in New York .-->
      <span> <target id="t44"/>.... <target id="t56"/> </span>
    </opinion_expression>
</opinion>
```

The span of the opinion expression matches the source entity *Zetsche* and event *say*. From this we derive a positive value for the attitude of the source towards the triples in its scope.

6.2 Factuality module

Currently, we have implemented a basic module that takes a few simple features into account based on modal adverbs and tense. We will develop a more elaborate version in the 3rd year of the project. We now first describe how factuality needs to modeled within the attribution module and summarize the main aspects that need to be considered when building a more elaborate factuality processor.

Event factuality is a property of the events expressed in a (written or oral) text. We follow Saurí (2008) conception of event factuality, which is "understood as the level of information expressing the commitment of relevant sources towards the factual nature of eventualities in text. That is, it is in charge of conveying whether eventualities are characterised as corresponding to a fact, to a possibility, or to a situation that does not hold in the world." The term eventualities is used here to refer to events, which can be processes or states. The main characteristics of events are that they have a temporal structure and a set of participants.

Factuality is not an absolute property, but it is always relative to a source, since events are always presented from the point of view of someone. The source does not need to be the author of a text, several sources can be reporting about the same event and the same source can assign different factuality values to an event along different points in time. Additionally, we assume that factuality value assignments are made at a specific point in time.

We also follow Saurí in considering three factuality components: source, time and factuality value. Source refers to the entity that commits to the factuality value of a certain event. Time is relevant because the factuality values of an event can change not only depending on the source but also along time. By default it will be the document creation time.

The factuality values will be characterised across two dimensions: polarity and certainty. The certainty dimension measures to which extent the source commits to the correlation of an event with a situation in the world, whereas the polarity dimension encodes whether the source is making a positive or a negative statement about an event happening in the world. The certainty dimension can be described as a continuum ranging from absolute certain to absolute uncertain. For the sake of simplicity, here we will conceive it as a discrete category with two values, certain and uncertain, instead of Sauri's three values (certain, probable, possible). Polarity is a discrete category which can have two values, positive or negative. Additionally both categories have also an underspecified value, for cases in which there is not enough information to assign a value. Events will be assigned one value per dimension.

Certainty	Polarity
certain (C)	positive (P)
uncertain (UC)	negative (N)
underspecified (U)	underspecified (U)

Table 30: Certainty and polarity values

Like in Saurí's work, the factuality profile will consist of four elements: an event in focus, a source, a factuality value per dimension, and a time, which is not implemented in her system. Additionally, we add an introducing source (isource) and an introducing time (itime) to refer to the source who is introducing another source and to the time in which this is happening. This is to account for source introducing predicates of the type *say*, *think*, etc.

Since we will be developing a rule-based system it is essential to analyse in detail the linguistic features that are involved in the expression of factuality. The system implements rules to account for all the features. We will follow Saurí's description, which is very complete and provides detailed information. It will be necessary to determine which are the faculty markers, how do they interact with each other, what syntactic constructions allow factuality markers scope over other markers, and what syntactic constructions block the scope.

A very summarised list of linguistic markers that assign factuality is provided below.

- Tense, aspect
- Lexical markers
  - Polarity markers: *no*, *nobody*, *never*, *fail*, ... They can act at different structural levels. At the clausal level they scope over the event-referring expression; at the subclausal level they affect one of the arguments of the event; at the lexical level by means of affixes. Polarity markers can negate the predicate expressing the event, the subject, the direct or indirect object.
  - Modality markers (epistemic or deontic) include verbal auxiliaries, adverbials and adjectives: *may, might, .perhaps, possible, likely, hopefully, hopeful, ...*
  - Commissive and volitional predicates (offered, decided) assign the value underspecified the subordinated event.
  - Event selecting predicates (ESP) are predicates that select for an argument denoting an event. Syntactically they subcategorise for a that-, gerundive or infinitival clause: *claim, suggest, promise, request, manage, finish, decide, offer, want, etc.* ESP project factuality information on the event denoted by its argument. Depending on the type of ESP they project different factuality values.

For example, *prevent* projects a counterfactual value, while *manage* projects a factual value. Some ESPs such as *claim*, are special in that they assign a factuality value to the subcategorised event and at the same time they express who is the source that commits to that factuality value, without the author of the statement committing to it. ESP can be source (SIP) or non-source introducing predicates (NSIP). SIPs such as *suspect* or *know* are ESPs that contribute an additional source relative to which the factuality of the subcategorised event is assessed.

- Some syntactic constructions can introduce a factuality value.
  - In participial adverbial clauses the event in the subordinated clause is presupposed as true (Having won Slovenia's elections, political newcomer Miro Cerar will have to make tough decisions if he is to bring stability to a new government).
  - In purpose clauses the main event is presented as underspecifed (Government mobilizes public and private powers to solve unemployment in the country).
  - In conditional constructions the factuality value of the main event in the consequent clause is dependent on the factuality of the main event in the antecedent clause ().
  - In temporal clauses the event is presupposed to be certain in most cases (While the main building was closed for renovation, the architects completed the Asian Pavilion).

  Additionally, some syntactic constructions act as scope islands, which means that the events in that construction cannot be affected by markers which are outside the constructions and at the same time the markers in the construction cannot scope over the events which are outside the construction.

  - Non-restrictive relative clauses (The new law might affect the house owners who bought their house before 2002).
  - Cleft sentences (It could have been the owner who replaced the main entrance door).

An event can be within the scope of one or more lexical markers, which means that in order to compute its factuality, the values of all the markers that scope over it have to be considered, as well as the ways in which markers interact with each other. It is also necessary to take into account that some syntactic contexts establish scope boundaries.

6.2.1 Factuality processor

In this subsection, we describe the factuality processor which should account for the phenomena described in the previous subsection. It concerns a rule-based system that relies heavily on syntactic information. The design of the system has to be based on deep linguistic analysis of the factuality phenomenon. It will be built in several phases:

1. Factuality values for verb events.
2. Factuality values for nominal events.
3. Factuality values and source identification.
4. Factuality values, source identification and time assignment.

Input – output

We assume that event tokens have been identified using other tools. The input for the processor is a tokenized document and the list of events in the document identified by token number. The output is the list of events with the assigned factuality values for each of the two factuality dimensions.

<factualitylayer >
<factvalue eid="t257" polarity="P" certainty="C" source="U" isource="U" time="U" itime="O"/>
<factvalue eid="t299" polarity="N" certainty="UC" source="t270" isource="U" time="2009-11-2" itime="2009-11-3"/>
<factvalue eid="t474" polarity="U" certainty="U" source="t470" isource="t460" time="U" itime="2009-XX-XX"/>
</factualitylayer>

eid is the token number that identifies the event.

polarity encodes the polarity values Positive, Negative, Underspecifed.

certainty encodes the certainty values Certain, UnCertain, Underspecifed.

source is the token number(s) that identify the source of the factuality judgement.

isource is the token number(s) that identify a source that has been introduced by another source.

time is the time at which the source has made the factuality statement.

itime is the time at which the introduced source has made the factuality assessment.

Preprocessing

The input for the preprocessing step will be the tokenised documents, the output will be the tokenised document with syntactic and semantic role information. The syntactic information is necessary to determine the scope boundaries, embedding of events and subordination structures, among others. Semantic role information is necessary to determine whether the full event or an event participant, which includes adjuncts, is affected by the factuality values.

The tokenised documents will be first preprocessed to obtain syntactic information. Since complex and reliable syntactic information is needed, we will experiment with several parsers (dependency and constituent, MaltParser and Stanford) in order to determine which one or which combination provides the most reliable and useful information. It might be the case that, depending on the syntactic features needed a type of parser is better than another or that the combination of two parsers produces more reliable information.

Lexicons

The factuality lexicons will be initially adapted from Sauri's lexicons. The lexicons encode information about the factuality values that lexical markers assign and about how do they interact with other markers. An example entry for the verb investigate is shown below. The *know_that* type assigns a certainty value to the event which is its complement and does not modify its polarity, whereas the *wonder* type assigns an underspecified value both for certainty and polarity.

Investigate	Complement:	comp	Factual type:	know_that
		dobj		know_that
		if comp		wonder

Table 31: Lexicon entry for investigate

6.2.2 Modules

Factuality will be processed in several steps, each of which will be a module of the system. As it has been indicated, events have been already identified and the documents have been preprocessed with syntactic parsers.

- Identification of lexical factuality markers and their types. This module will access the lexicons and will output the token number of the factuality marker and its type. The module needs to have access to syntactic information in order to identify the syntactic structures in which the markers occur. In some cases the same marker might belong to different types depending on the syntax. Some word sense disambiguation might be needed, since some factuality markers have more than one sense (for example the modal auxiliary *can*).

- Identification of the syntactic constructions in which events are embedded in order to determine whether the event is embedded in a construction that affects the factuality values. The output of this module is the event identifier and one or more syntactic constructions.

- Determining scope boundaries per event in order to determine how much context from the event token to the beginning of the sentence has to be analysed to find factuality markers that scope over the event. The output should be the event identifier and the identifier of the token where the search for factuality markers should stop or the list of token identifiers that do not have to be included in the markers search.

- Assigning factuality values to events dynamically. With the information from the previous modules, this module should compute the final factuality value for each event. It will iterate through the factuality markers that are to the left of the event (in the word sequence order, for English) respecting scope boundaries and it will update the factuality value of the event accordingly, marker by marker, taking also into account the constraints of the syntactic context of the event.

- Identifying source and isource per factuality marker.

- Identifying time per factuality marker.

6.2.3 Evaluation

The preprocessing modules will be evaluated with standard datasets (from CoNLL shared tasks for example). Every module of the factuality processor will be evaluated independently with gold standard data. The full factuality processor will be evaluated with the FactBank corpus and with a manually annotated small dataset developed specifically for this task.

6.2.4 Multilinguality

The factuality processor will be developed initially for English. In order to adapt the module to other languages new factuality lexicons will be needed and the syntactic rules to find the scopes of factuality markers will have to be adapted. The lexicons can be probably translated from the English lexicons and manually revised by a linguistics expert in order to check whether the same factuality behaviour applies. More costly might be to adapt the syntactic rules, though part of the cost can be reduced by preprocessing the documents with parsers that have models for several languages (such as MaltParser). We will make a predevelopment analysis of the cost of adapting the processor to other languages in order to design it in such a way that the cost can be maximally reduced.

A first version of the factuality module for English has been implemented for the SemEvel 2015 timeline extraction task. In the third year of the project, we will develop further releases of this module for English and the other languages.

7 Conclusions

In this deliverable we reported the results on event modeling achieved during the second year of the NewsReader project, as part of WP5 activities.

First, we presented a new "bag of events" approach to perform cross-document event coreference that takes document-based event templates as a starting point. The performances of the approach were evaluated against the ECB+ and ECB corpora, where it outperformed related work and other approaches (e.g., mention-based approach). The event coreference approach is implemented as part of the the NAF2SEM module which reads the NAF files with mention representations of entities and events and creates a RDF-TRiG representation with unique instances for entities and events pointing to the mentions on which they are based. The module successfully processed the new version of the CAR data set consisting of 1.26 million NAF files, producing approximately 440M RDF triples.

Second, we described our modules for extracting event relations, one for temporal relations and one for causal relations. Preliminary evaluations of both modules showed that their performances are comparable or even better that the state-of-the-art (the temporal relation extraction outperform all systems that participated to the TempEval-3 task at SemEval 2013).

Third, we reported our preliminary work on the construction of timelines, a first step towards creating storylines. While the initial results are not as satisfying as expected, we identified several strategies to improve our work, to be investigated in the next version of the timeline module.

Fourth, we reported some initial results on the cross-lingual processing of news document, obtained by comparing generated RDF-TRiG files for the Wikinews corpora for English, Spanish and Dutch.

And, to conclude, we presented a preliminary implementation of the perspective and attribution module, sketching the plan for its evaluation and its adaptation to a multilingual scenario.

The activities of the third year of the project on event modeling will focus on several streams of work. Among them, we plan to improve the event-coreference module using the benchmark results: in particular, we wil perform a further error analysis of the ECB+ and Wikinews corpus to finetune the current approach. To improve the timeline extraction module, we plan to use the benchmark results of the SemEval timeline task by tracing errors to the components that contribute negatively to the performance of the approach. This feedback will be used to improve these components or developed alternative solutions. We will also implement a direct interaction of the NAF2SEM module with the KnowledgeStore: related events are retrieved directly through SPARQL requests to the KnowledgeStore, and the results of the comparison are fed back to the KnowledgeStore. Furthermore, we plan to improve the attribution and perspective module, and to perform further experiments on the cross-lingual processing of news on the basis of the preliminary results here reported.

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